



MMS050 Finite Element Simulation In Design

Hand in Assignment 2

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Chalmers University of Technology
MMS050 Finite Element Simulation In Design

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1 Task 1: Design of a water tower

1.1 Sub Task

- The ends of the water tower are fixed supports. The thickness of the water tank is considered as 0.125m. The force is calculated including the mass of the water and the mass of the container. The force is acting downwards. An internal pressure load was applied to the tank walls.
- The material of container used is Structural steel. The thickness of the wall considered is 0.0125m. The density of water and other properties of structural steel is considered.

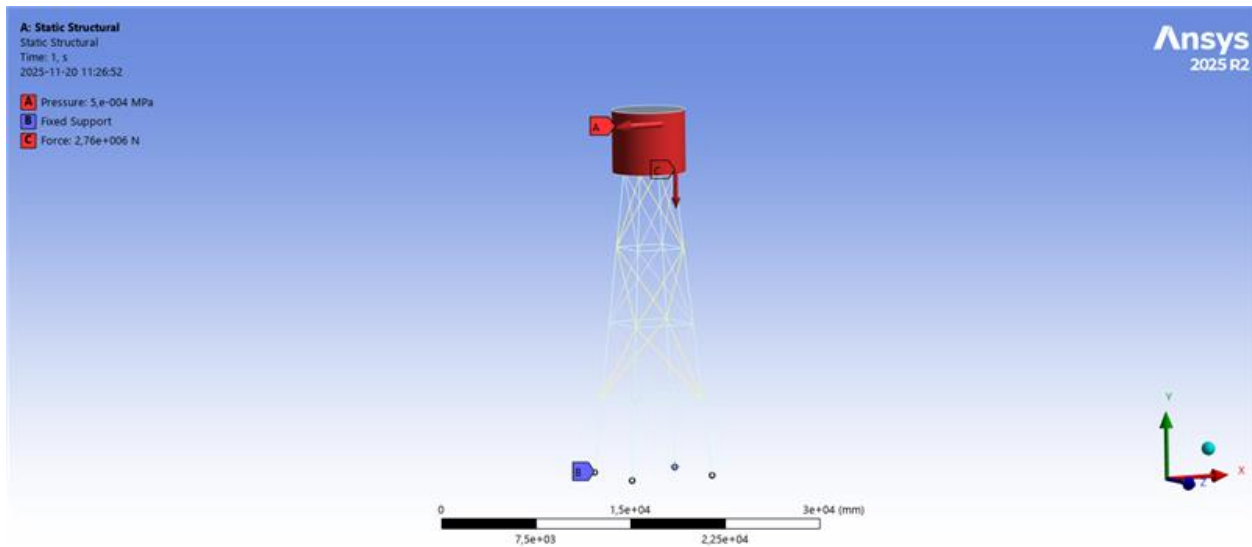


Figure 1: Boundary condition

1.2 SubTask

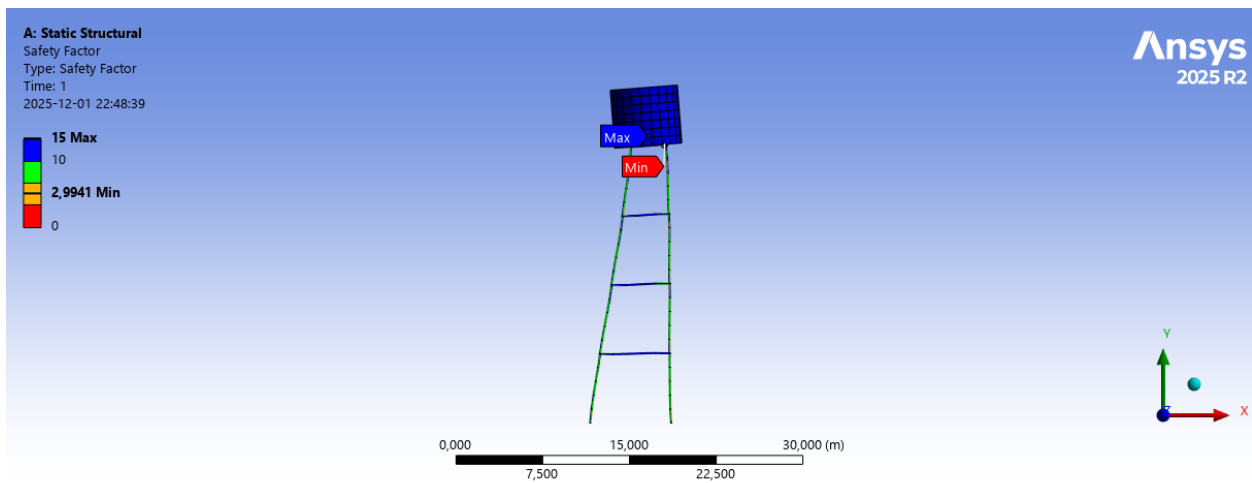


Figure 2: Factor of Safety

The Factor of safety is 2.99. The FOS is on the connection point at the top of the structure joining the tank and the tower.

1.3 SubTask

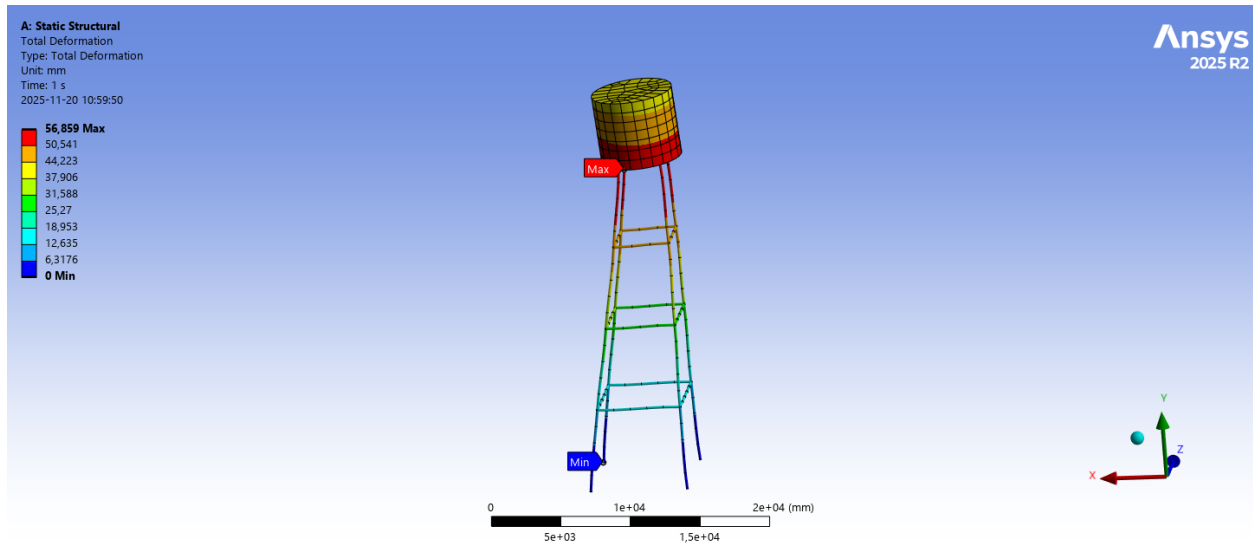


Figure 3: Total Deformation

The maximum deformation is 56.8 mm which is less than the threshold 200mm.

1.4 SubTask

Details of "Beam (Circular Tube)"	
Length Y	24, m
Length Z	7, m
Properties	
☐ Volume	3,4573 m ³
☐ Mass	27140 kg
☐ Length	168,66 m
☐ Cross Section Area	2,0498e-002 m ²
☐ Cross Section IYY	5,9055e-005 m ² ·m ²
☐ Cross Section IZZ	5,9055e-005 m ² ·m ²
Statistics	
Nodes	268
Elements	140
Mesh Metric	None

Figure 4: Total mass of the Structure

The total mass of the frame structure is 27140kg excluding the weight of the water container.

1.5 SubTask

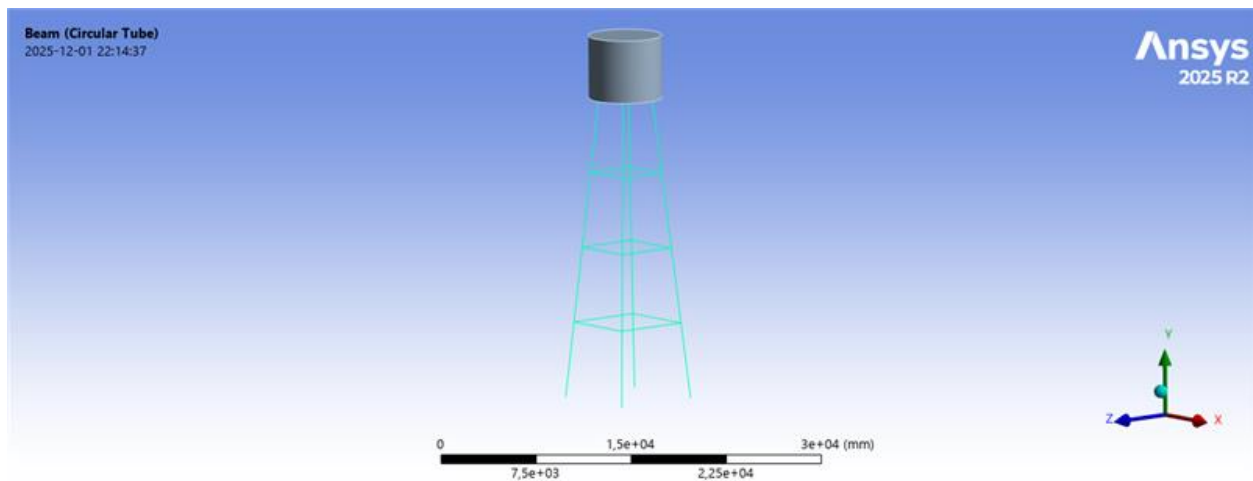


Figure 5: Total mass of the Structure

The cross section of the circular tube Outer radius is 0.095m, Inner radius is 0.05m.

2 Task2: A dam structure

2.1 SubTask

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The max princile stress is 2.758 MPa and Min principle stress is 3.831 MPa.

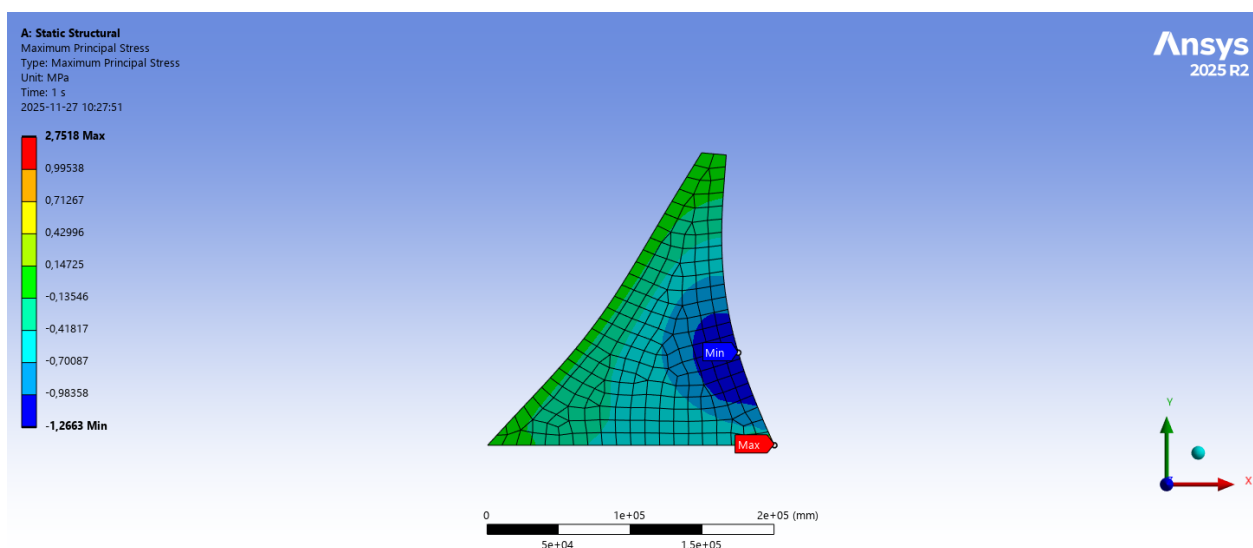


Figure 6: Max Principle stress for 2D

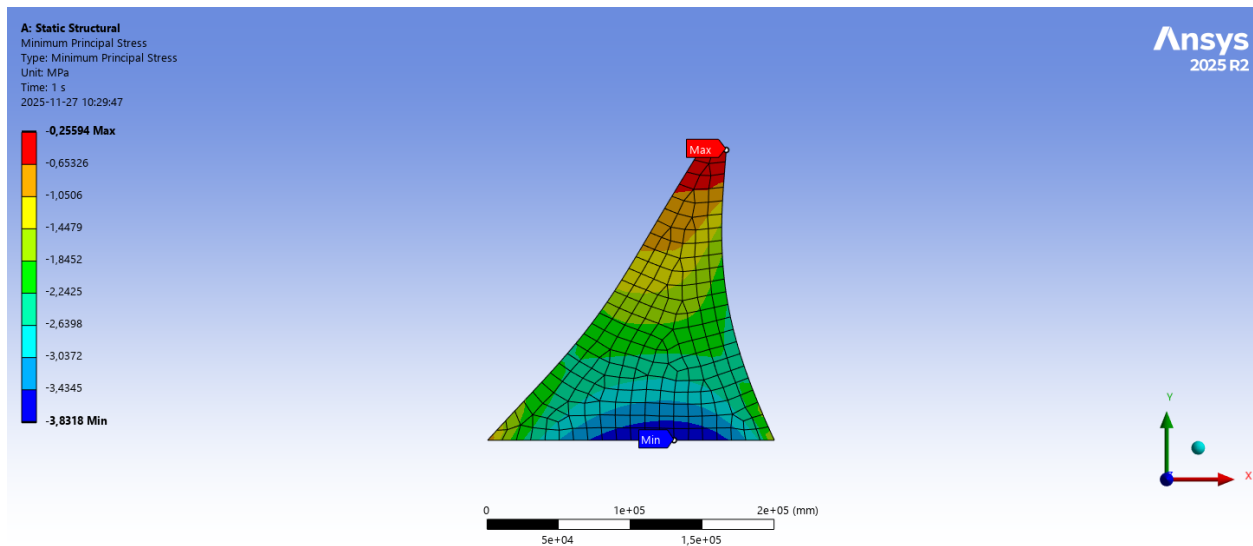


Figure 7: Min Principle stress for 2D

2.2 SubTask

The Section Plane is taken in a such way that when we view from side it is similar to the 2D geometry

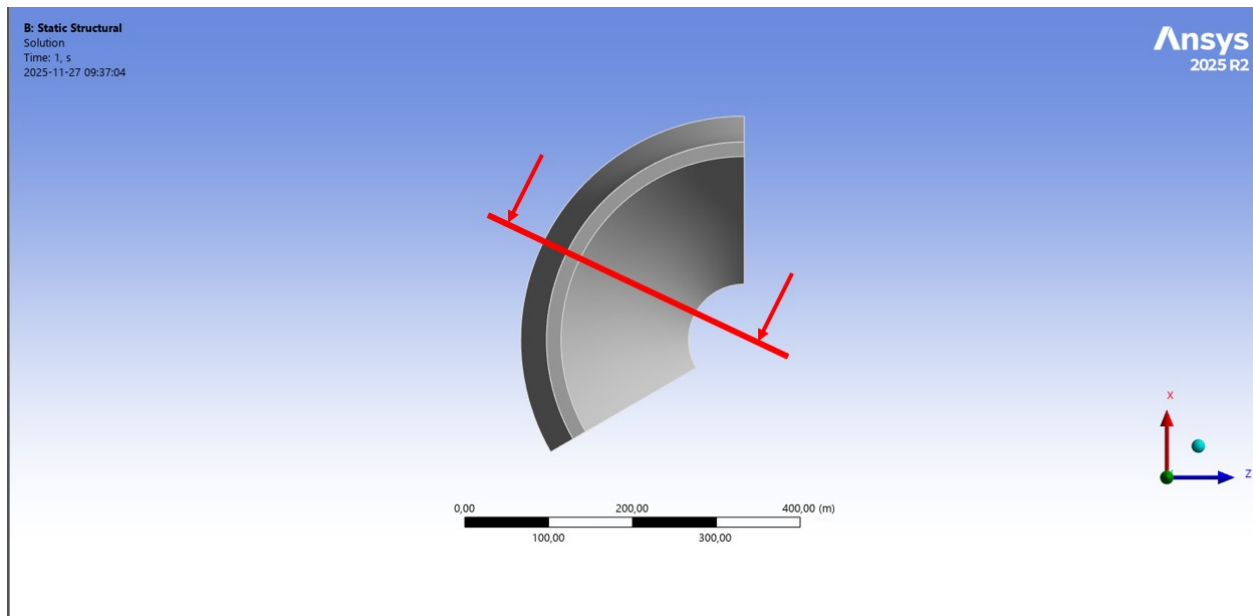


Figure 8: Cross section of Dam structure

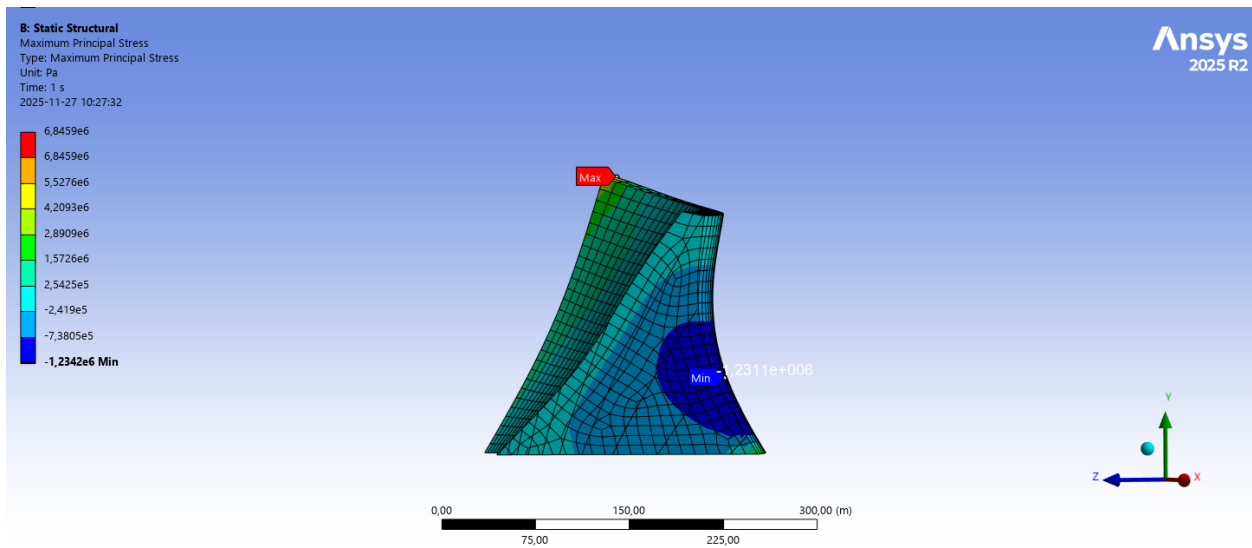


Figure 9: Max Principle Stress for 3D

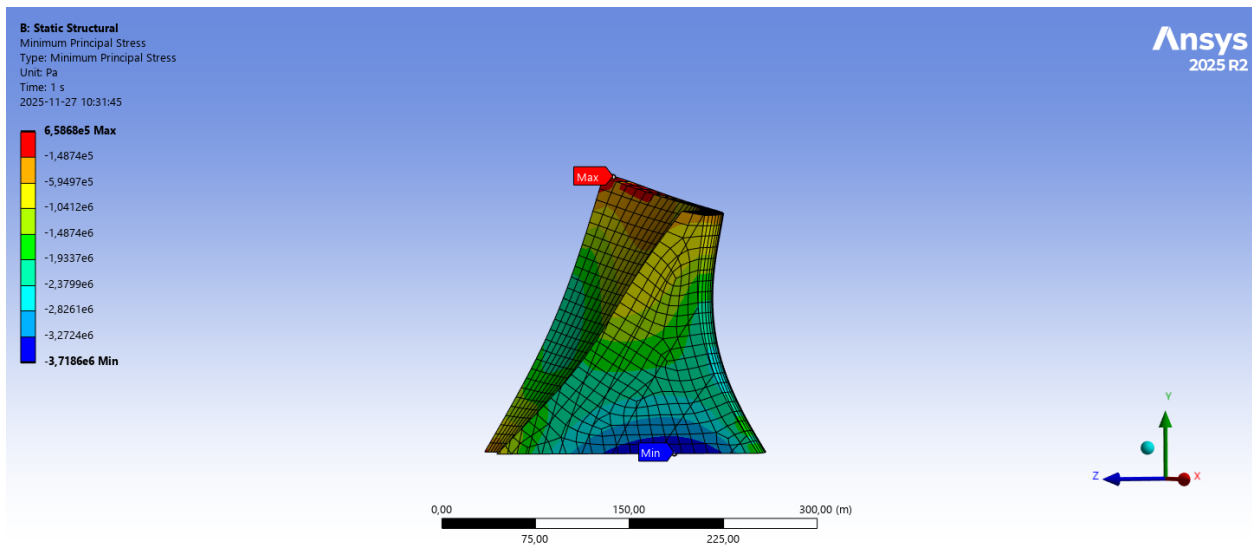


Figure 10: Min Principle stress for 3D

The Max and the min principle stress in 3D is almost same to 2D geometry as we try to measure using the probe in similar boundaries as of 2D. The max stress in 3D can be neglected because its pointing at the top edge corner of the Dam in 3D.

3 Task3: Meshing & Defeaturing

3.1 SubTask

- Default mesh

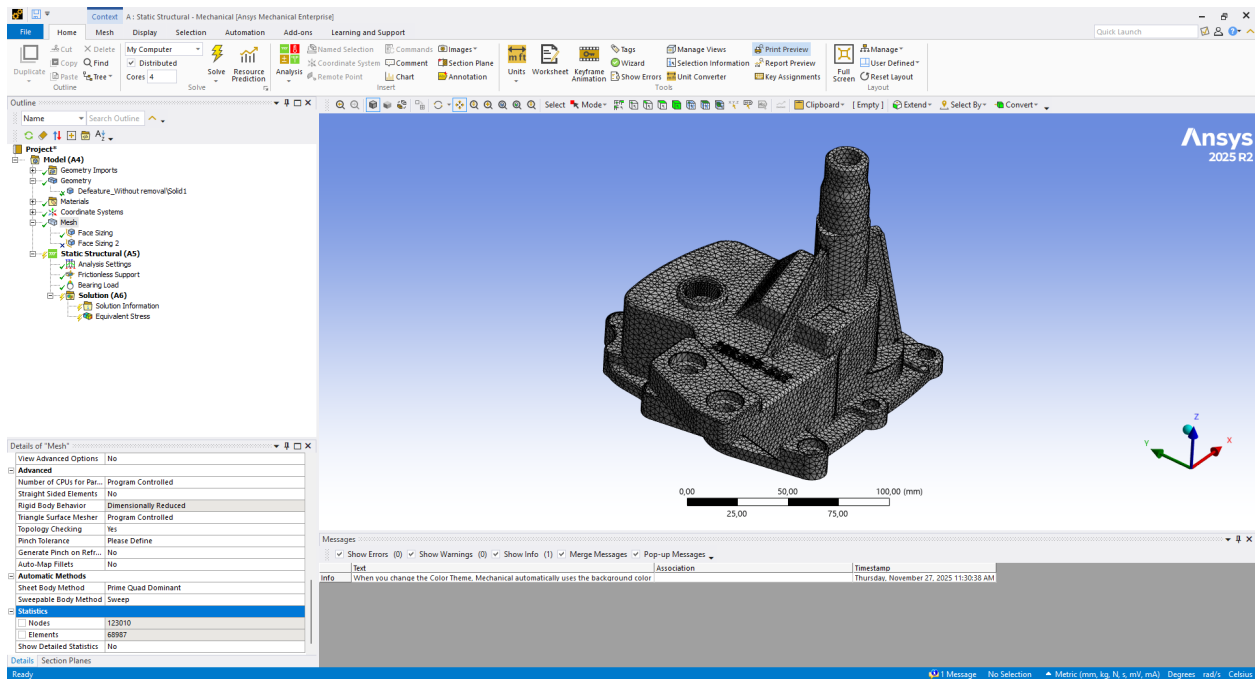


Figure 11: 3mm mesh size

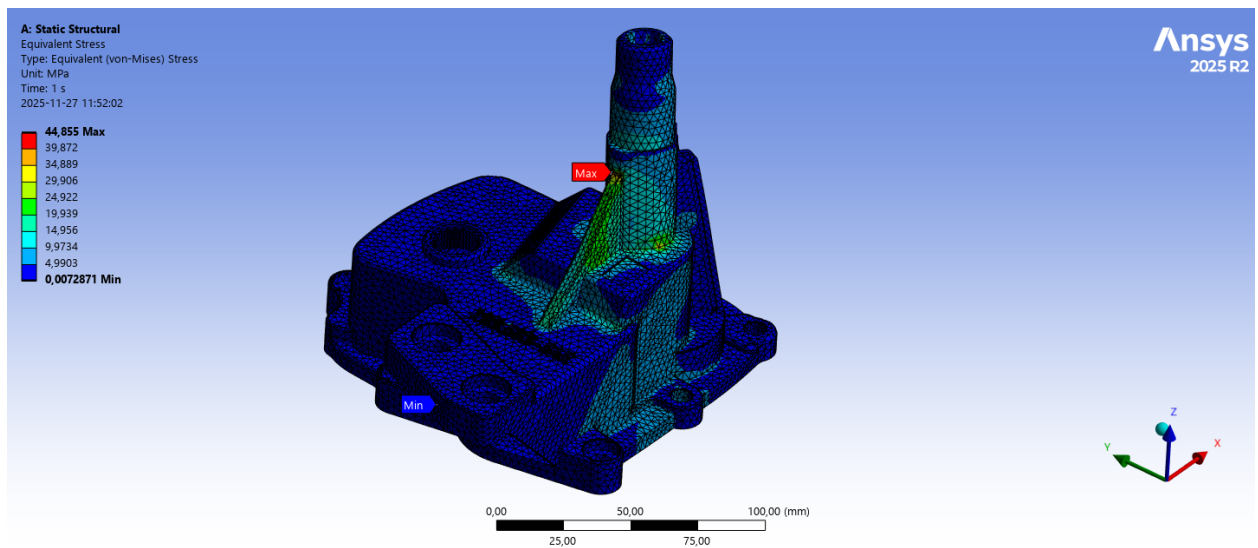


Figure 12: 3mm mesh size and max stress is 44.855MPa

The No. of Nodes is 123010 and No of Elements is 68987. The max stress at radius obtained is 44.855MPa.

- Coarse mesh with Refinement

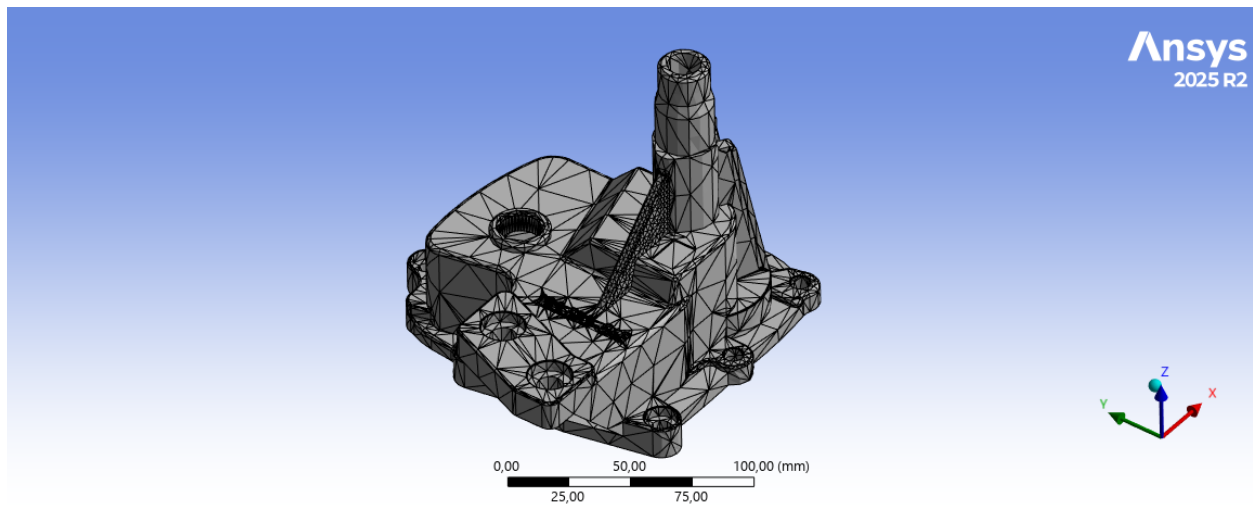


Figure 13: Coarse mesh with Refinement

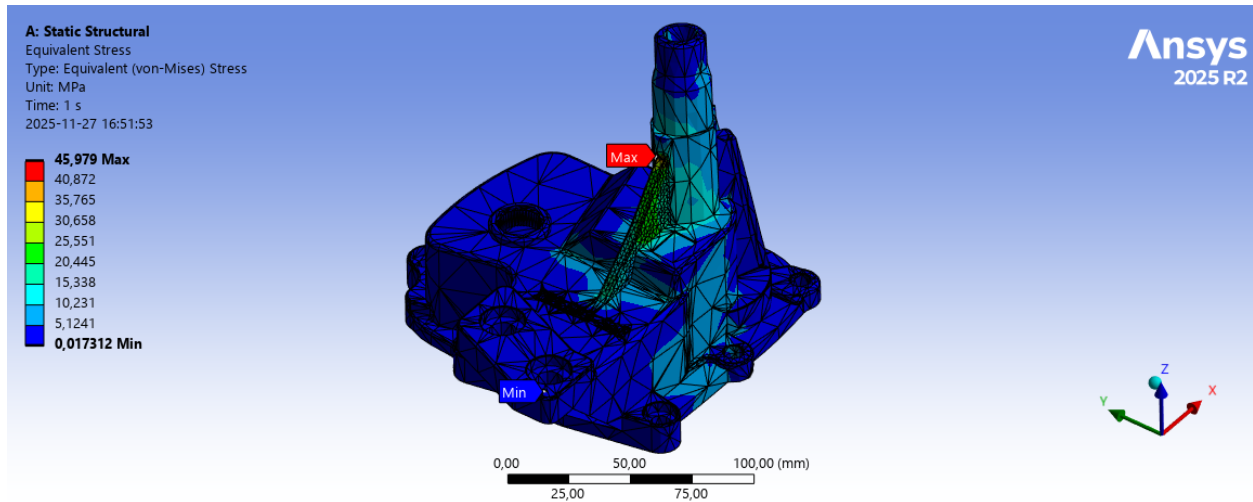


Figure 14: Coarse mesh with Refinement and Max stress is 45.97 Mpa

The No. of Nodes is 25498 and No of Elements is 13645. The max stress is at the radius and the value is 45.97 Mpa is observed with a reduction in mesh elements and nodes.

3.2 SubTask Defeaturing

- Default mesh

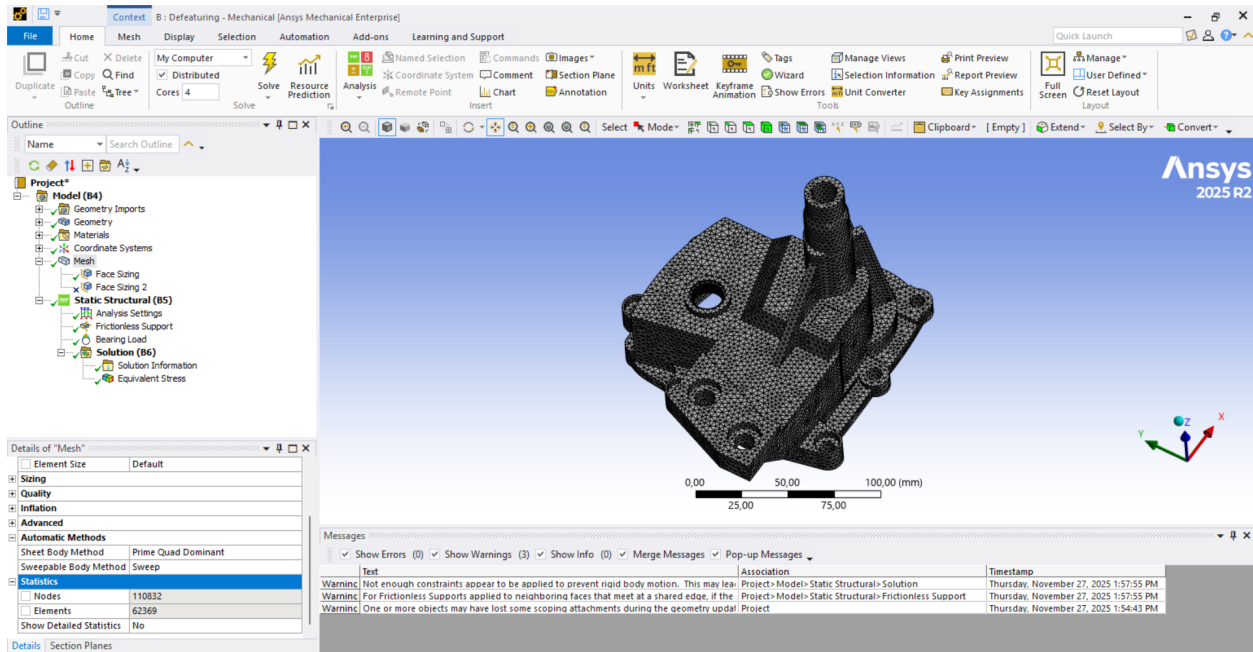


Figure 15: Defeating 3mm mesh size

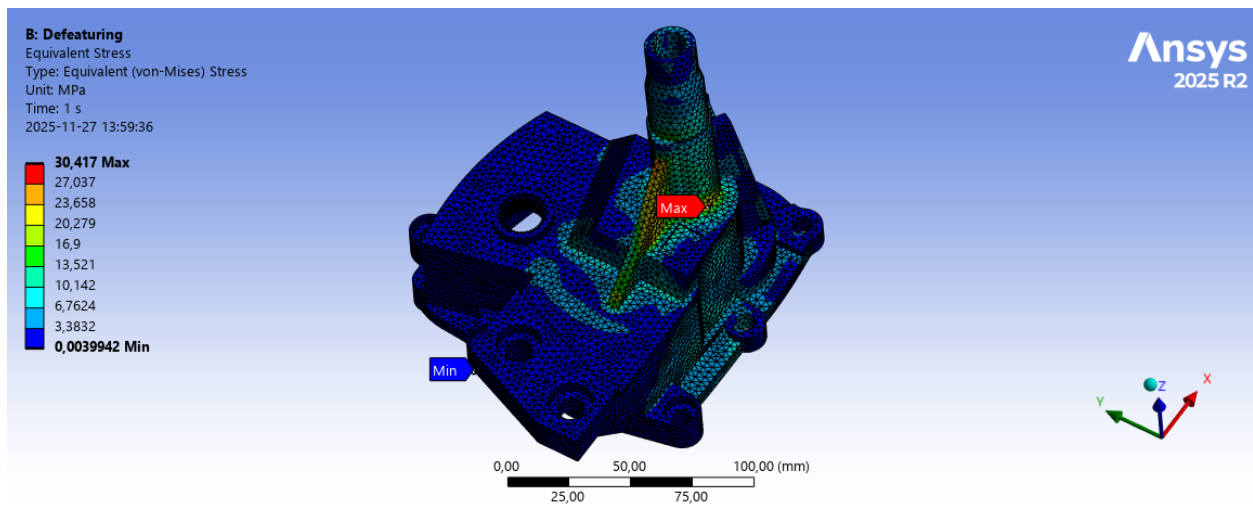


Figure 16: Defeating 3mm mesh and Max Stress is 30.417MPa

The Defeating model has No. of Nodes is 110832 and No of Elements is 62369. The max stress is at the radius and the value is 30.417 MPa.

- Coarse mesh with Refinement

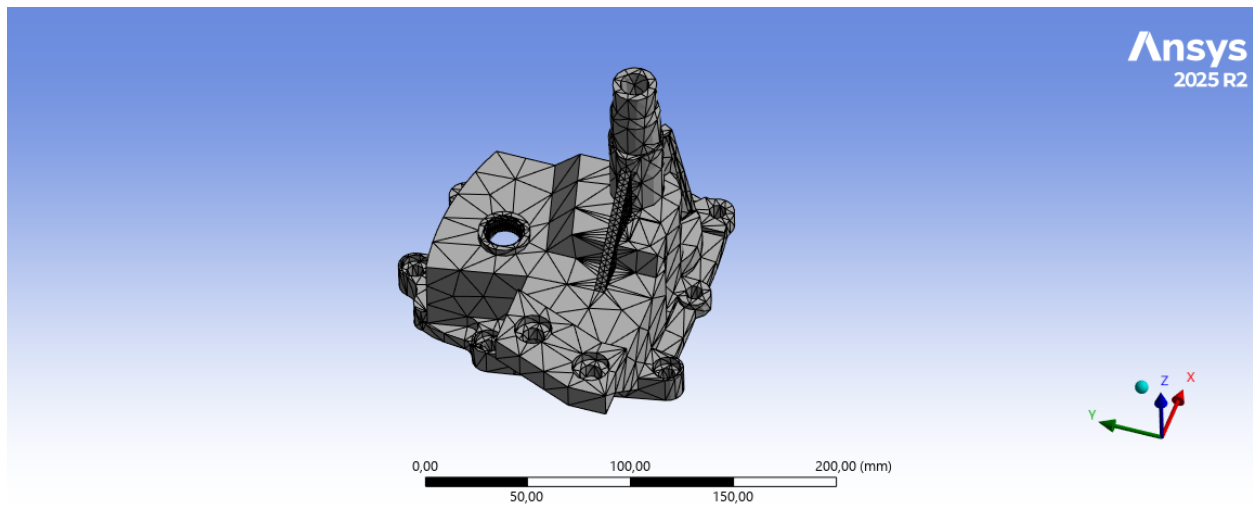


Figure 17: Defeaturing Coarse mesh with Refinement

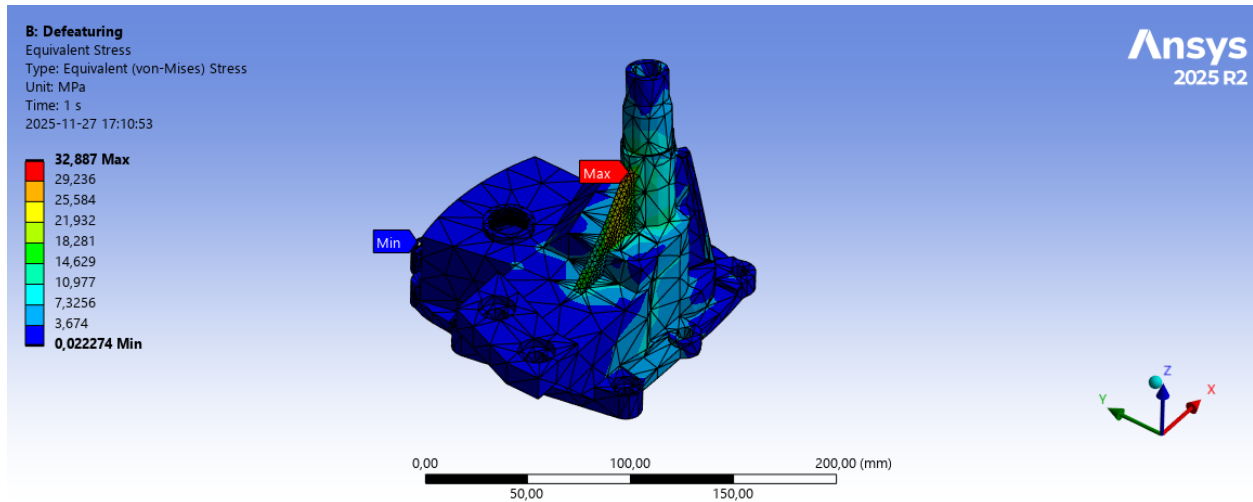


Figure 18: Defeaturing Coarse mesh with Refinement and Max Stress 32.887 MPa

The Defeaturing model with coarse mesh has No. of Nodes is 10045 and No of Elements is 5030. The max stress is at the radius and the value is 32.887 Mpa is observed with a reduction in mesh elements and nodes.

Conclusion: The coarse mesh with local refinements provides a good balance between accuracy and computational effort. Further defeaturing simplifies the geometry, reduces the number of elements, and still delivers comparable results, making the simulations more efficient without sacrificing key response trends.

4 Task 4: Bracket

4.1 SubTask

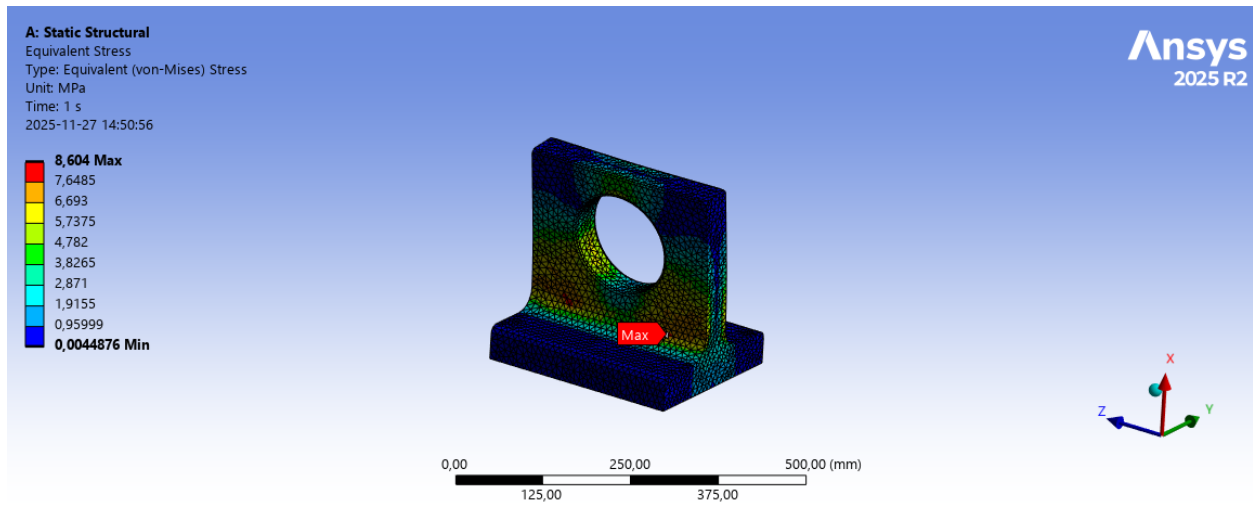


Figure 19: Maximum stress location for 10mm mesh

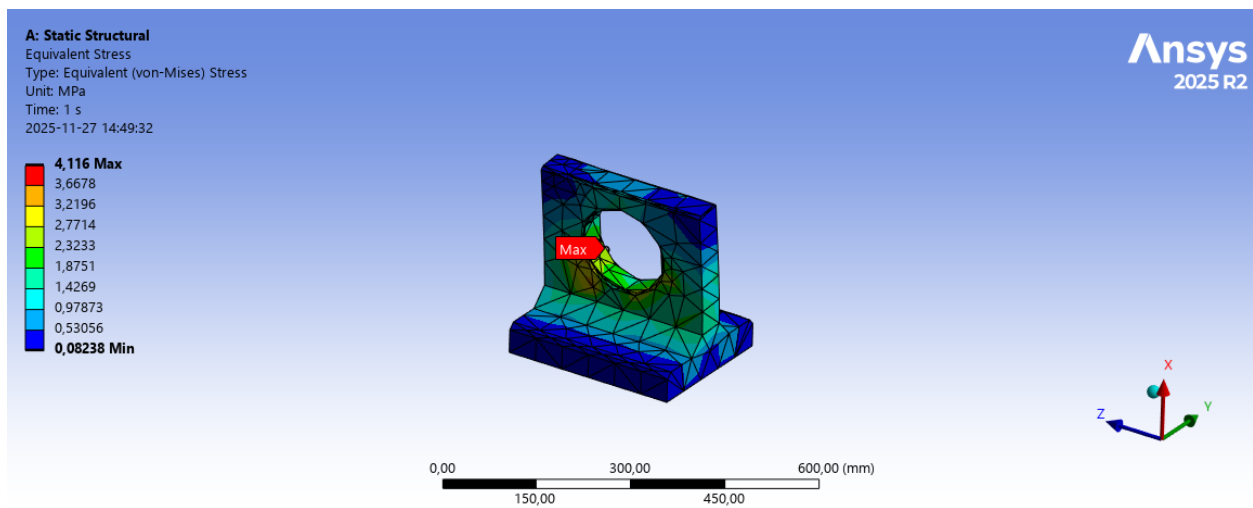


Figure 20: Maximum stress location for 50mm mesh

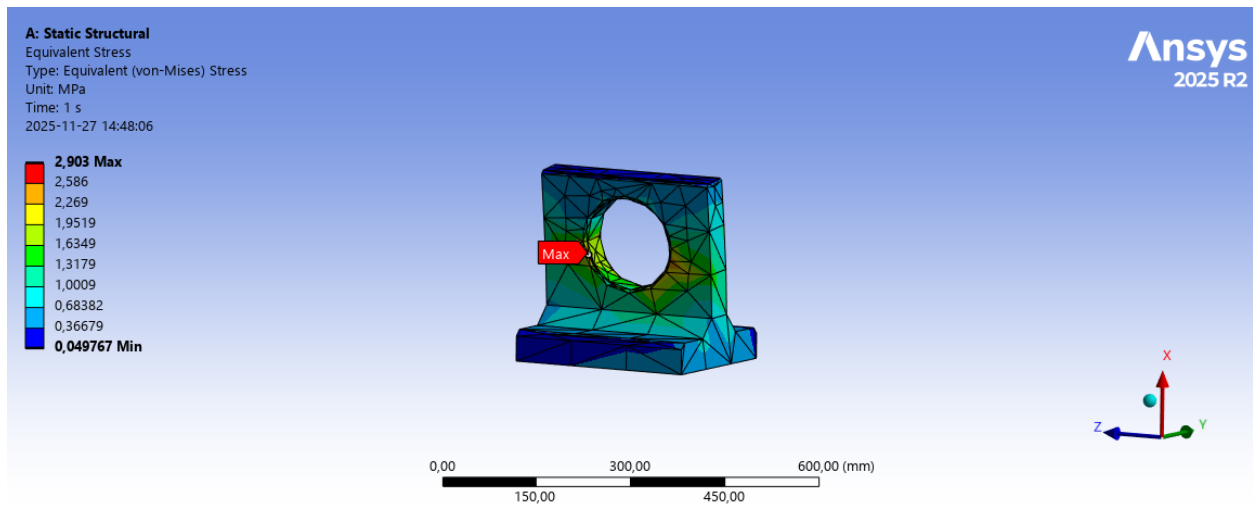


Figure 21: Maximum stress location for 100mm mesh

Conclusion: The finest mesh captured the stress location and magnitude much more accurately than the coarse mesh, highlighting clear differences in the results. This underlines how crucial mesh quality is for obtaining reliable and meaningful simulation outcomes.

4.2 SubTask

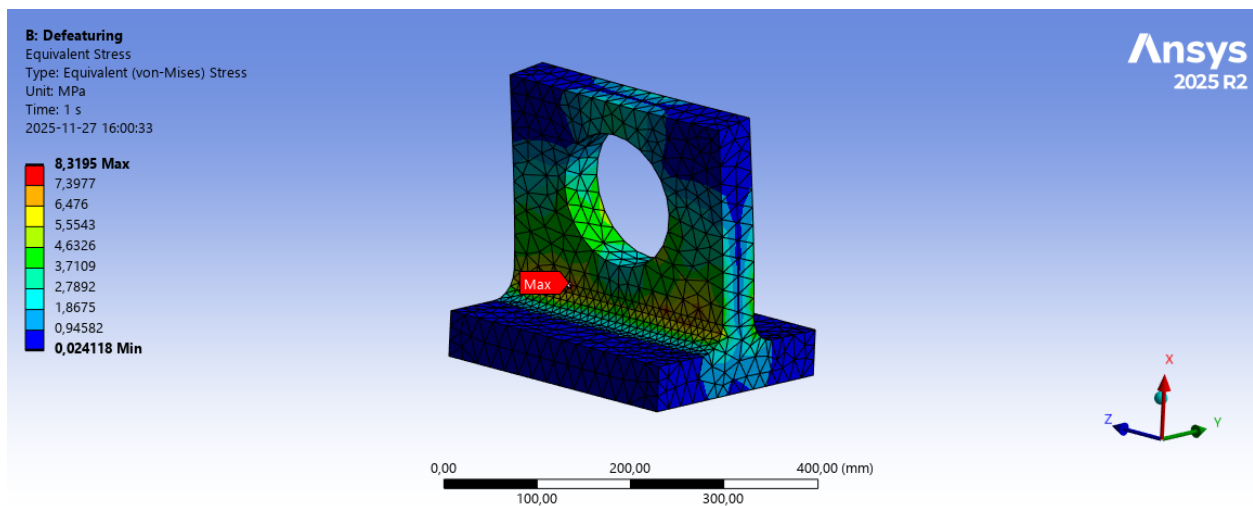


Figure 22: Maximum stress location for defeature 10mm mes

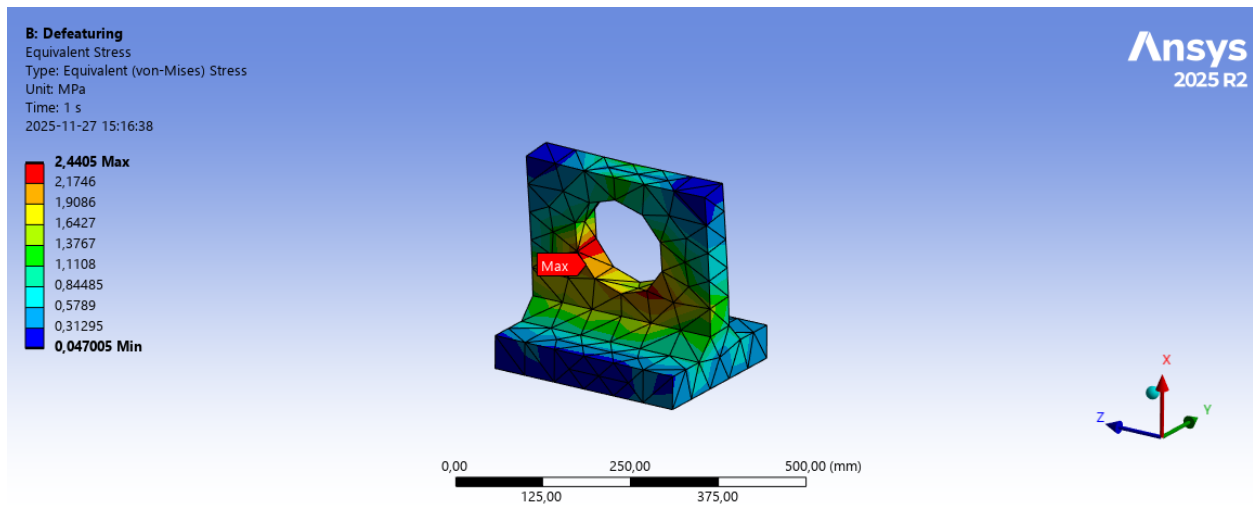


Figure 23: Maximum stress location for defeature 50mm mesh

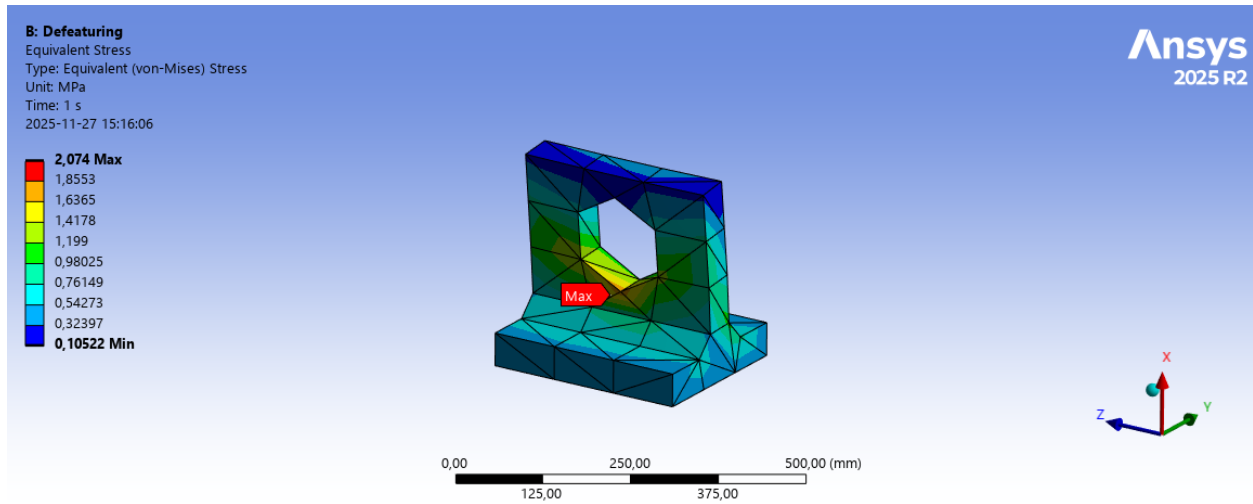


Figure 24: Maximum stress location for defeature 100mm mesh

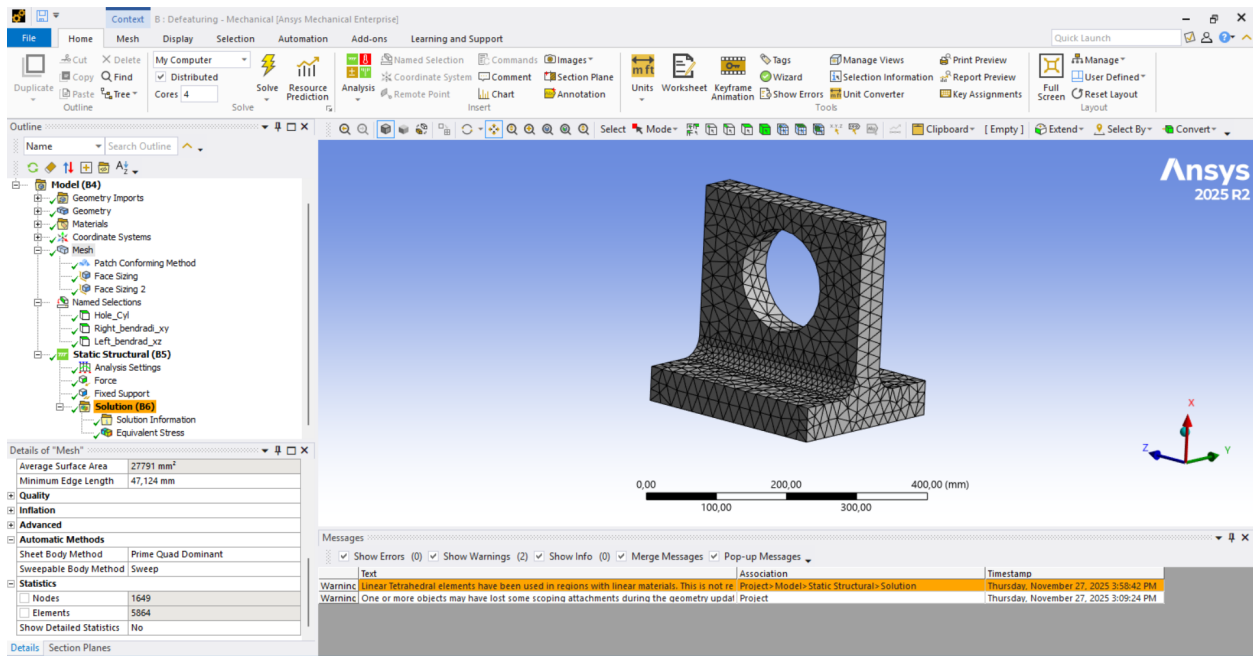


Figure 25: Mesh 10mm at fillet and 20mm mesh at rest of bracket

For this mesh size 10mm for fillet and 20mm mesh size for rest of the bracket, the No of nodes obtained is 1649 and No of elements are 5864.

5 Task 5: Space frame

5.1 SubTask

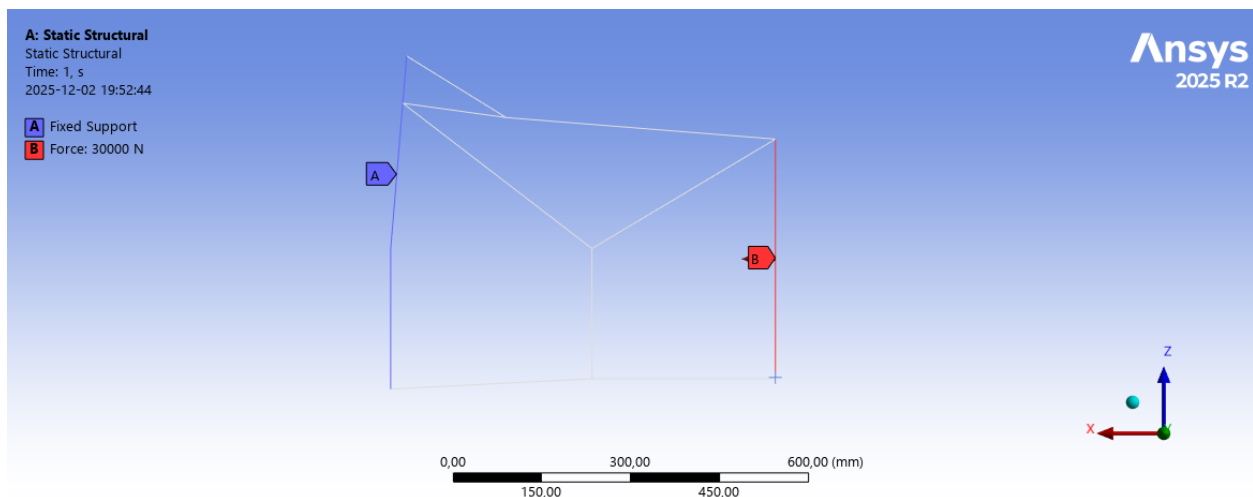


Figure 26: Boundary condition

For boundary condition i have applied the fixed support for the font hoop and the force of 30KN for the plane of impact.

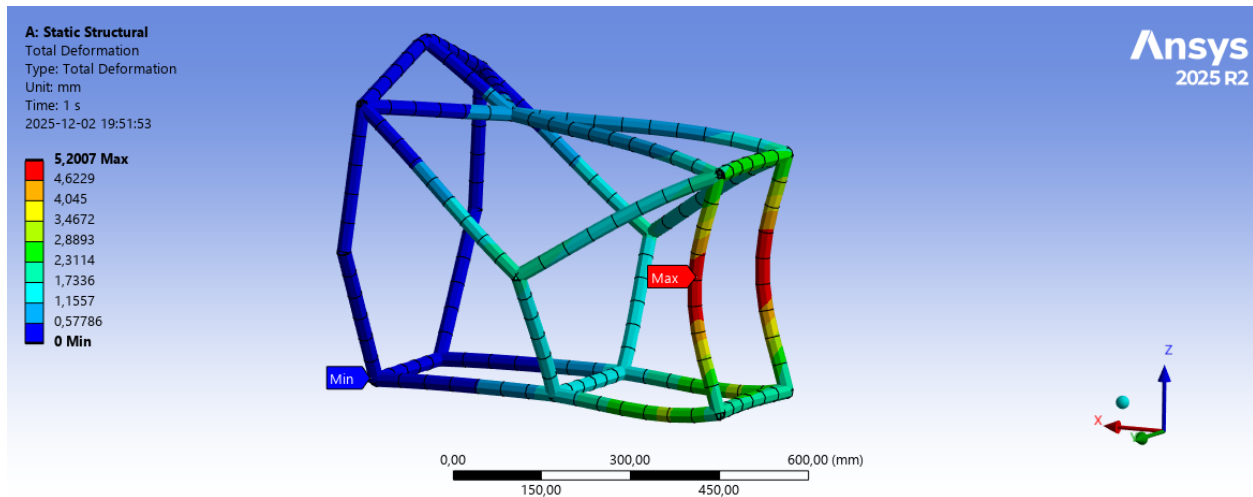


Figure 27: Total Deformation

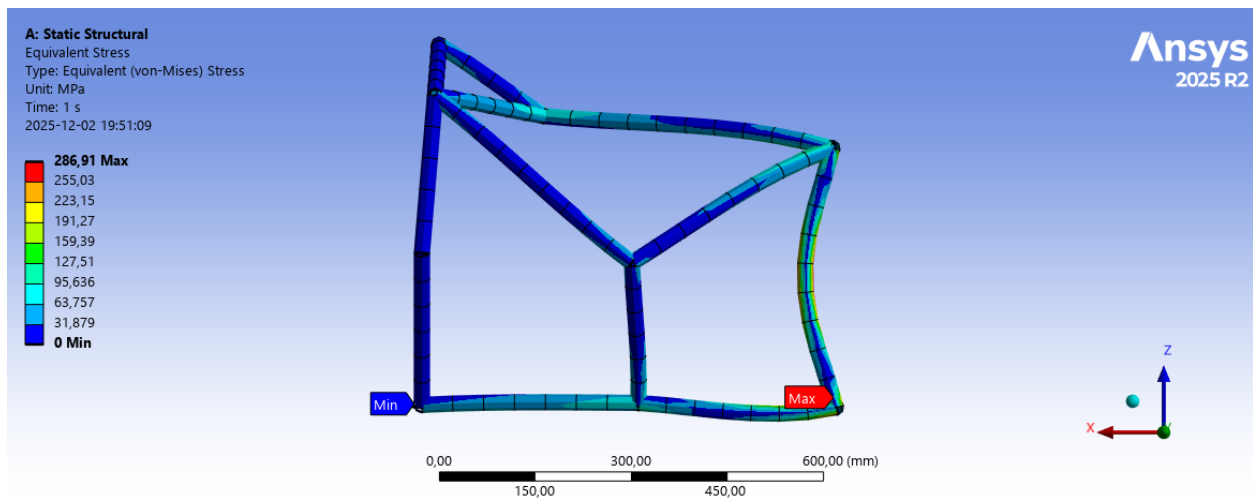


Figure 28: Max stress at plane of impact

The aluminium alloy of cross section OD $\phi 25.4\text{mm}$ and ID $\phi 20.57\text{mm}$ the total deformation is 5.2007 mm and max stress is 286.91 MPa shown above. The mass of the aluminium alloy frame is 4.0966Kg.

5.2 SubTask: Proposal 1

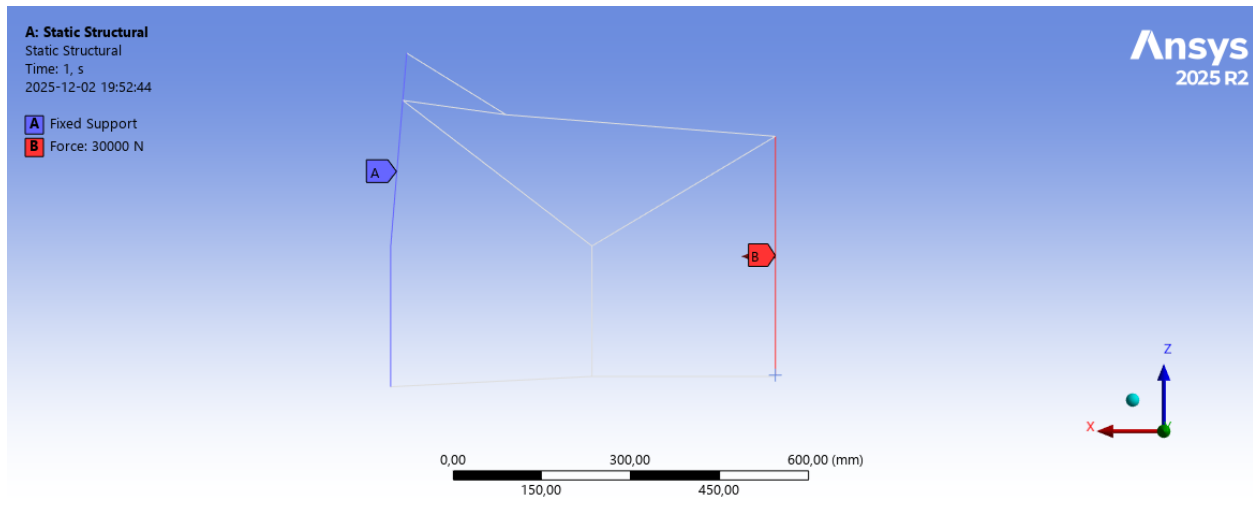


Figure 29: Boundary condition

For boundary condition i have applied the fixed support for the font hoop and the force of 30KN for the plane of impact.

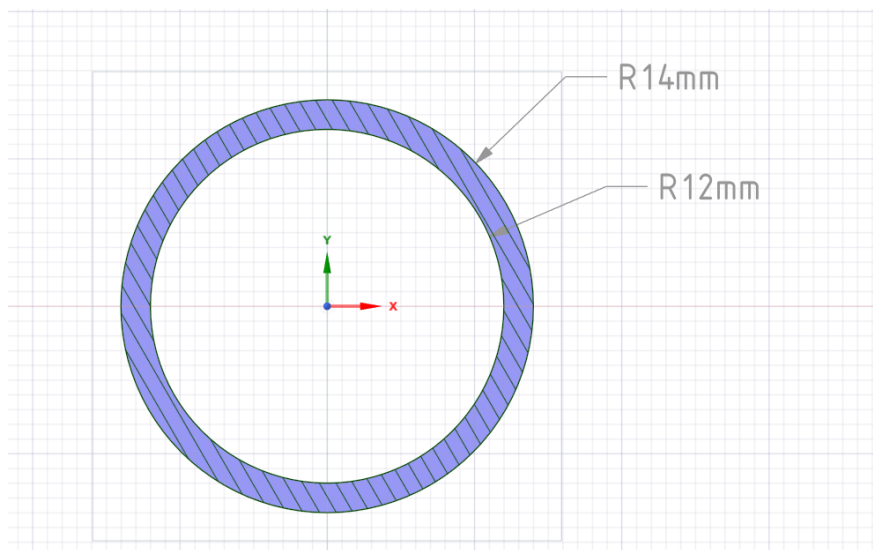


Figure 30: Cross section of aluminium alloy

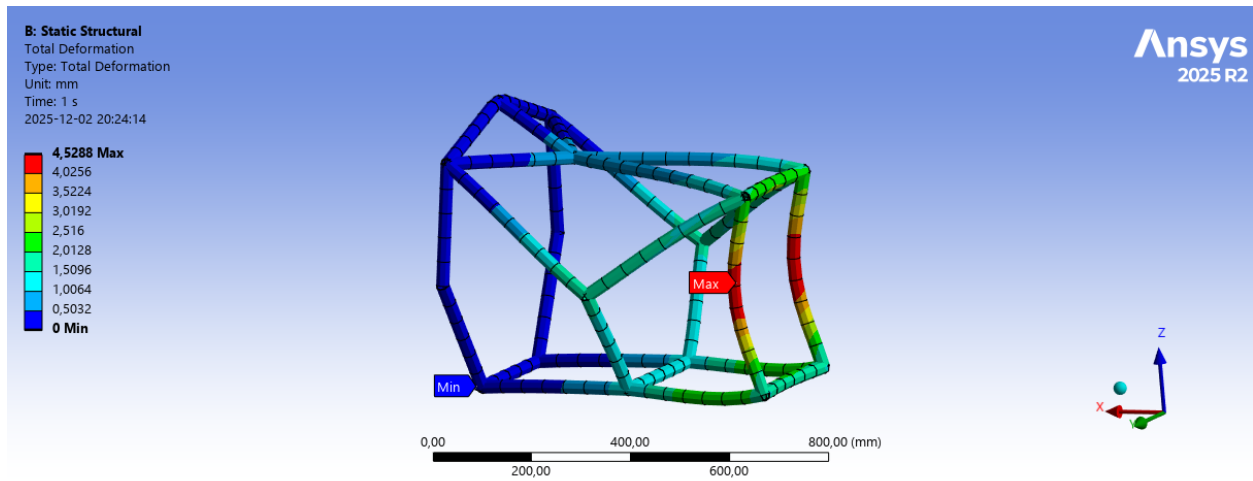


Figure 31: Total Deformation

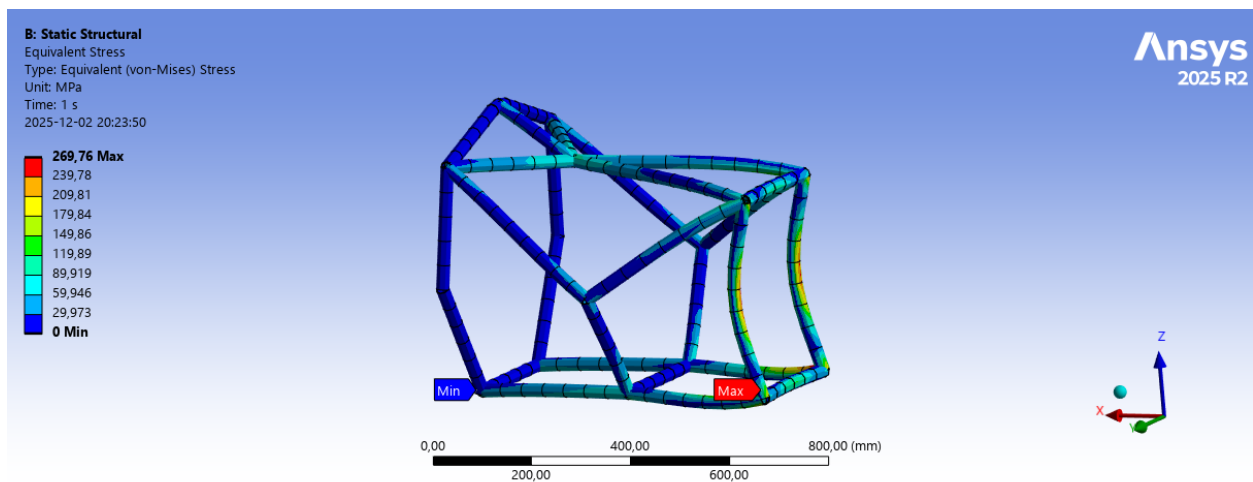


Figure 32: Max stress at plane of impact

The aluminium alloy of cross section OD $\phi 28\text{mm}$ and ID $\phi 24\text{mm}$ the total deformation is 4.5288 mm and max stress is 269.76 MPa shown above. The mass of the aluminium alloy frame is 3.8376Kg.

5.3 SubTask: Proposal 2

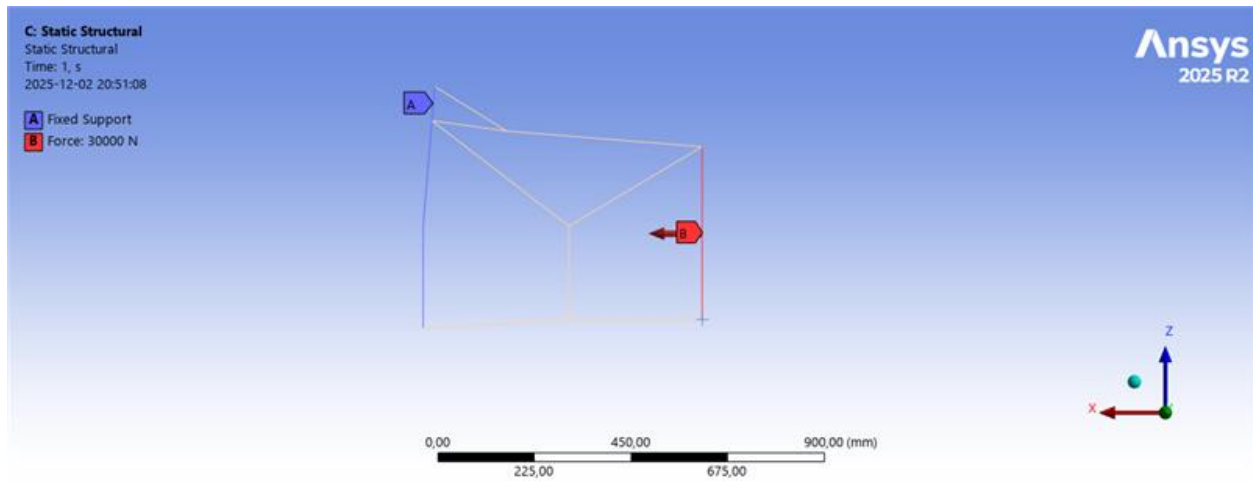


Figure 33: Boundary condition

For boundary condition i have applied the fixed support for the font hoop and the force of 30KN for the plane of impact.

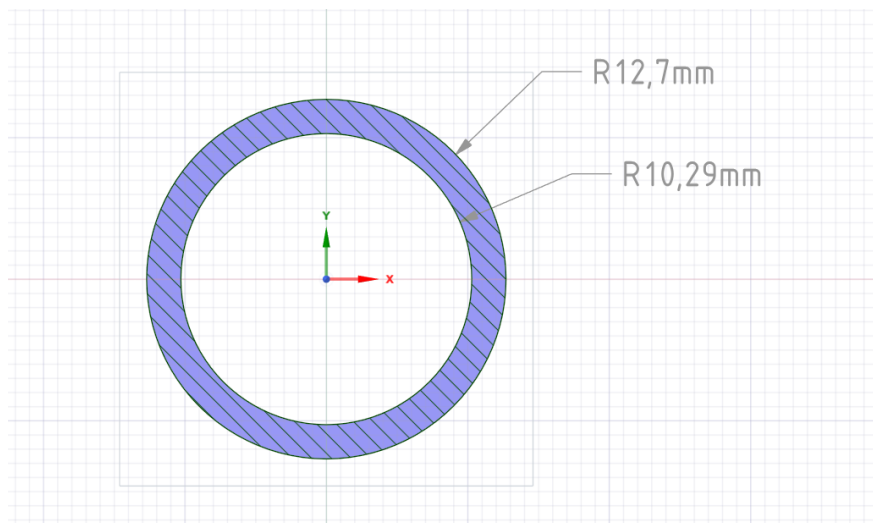


Figure 34: Cross section of aluminium alloy at plane of impact member

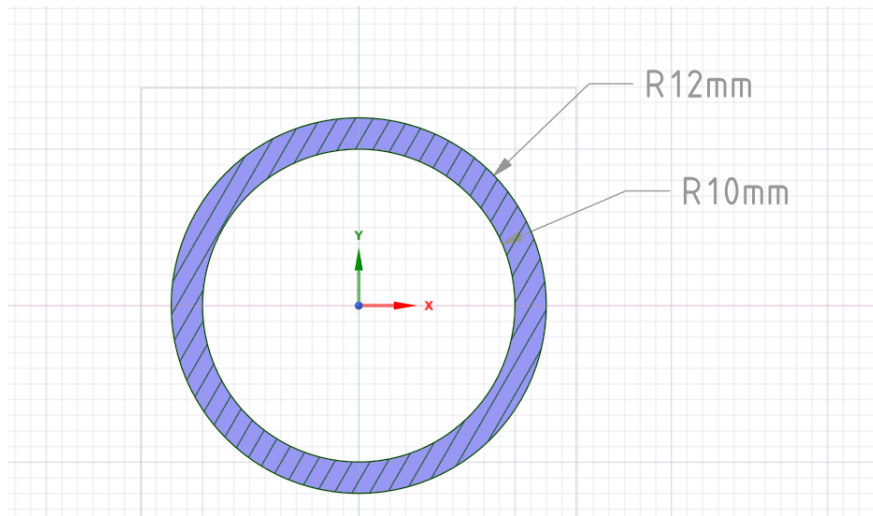


Figure 35: Cross section of aluminium alloy at front hoop member

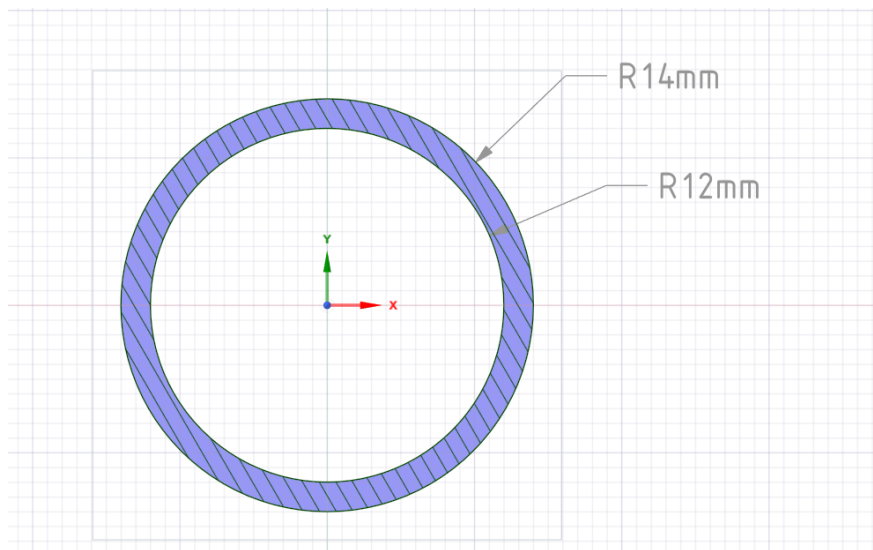


Figure 36: Cross section of aluminium alloy at rest of the member

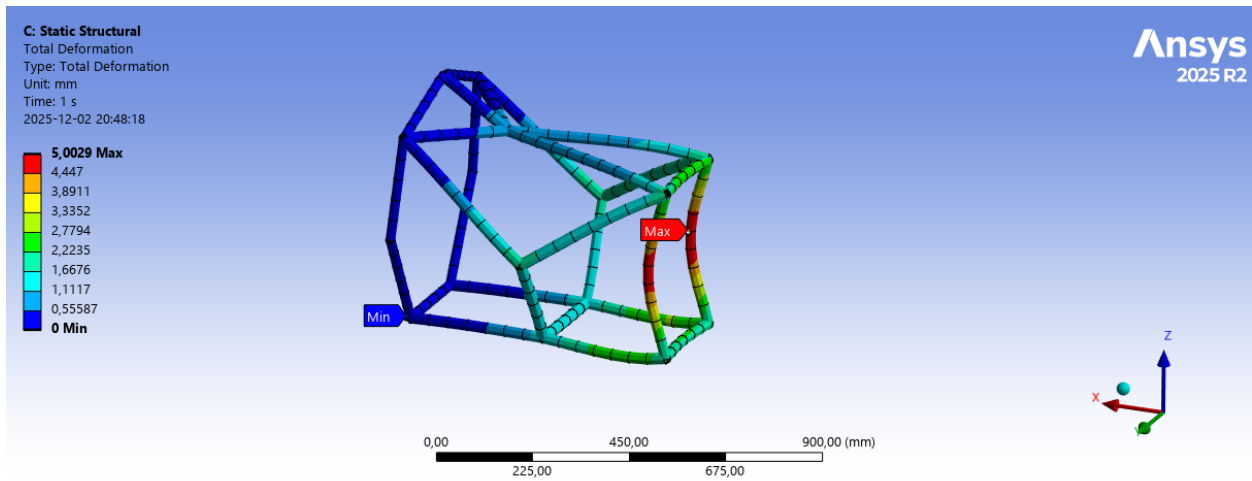


Figure 37: Total Deformation

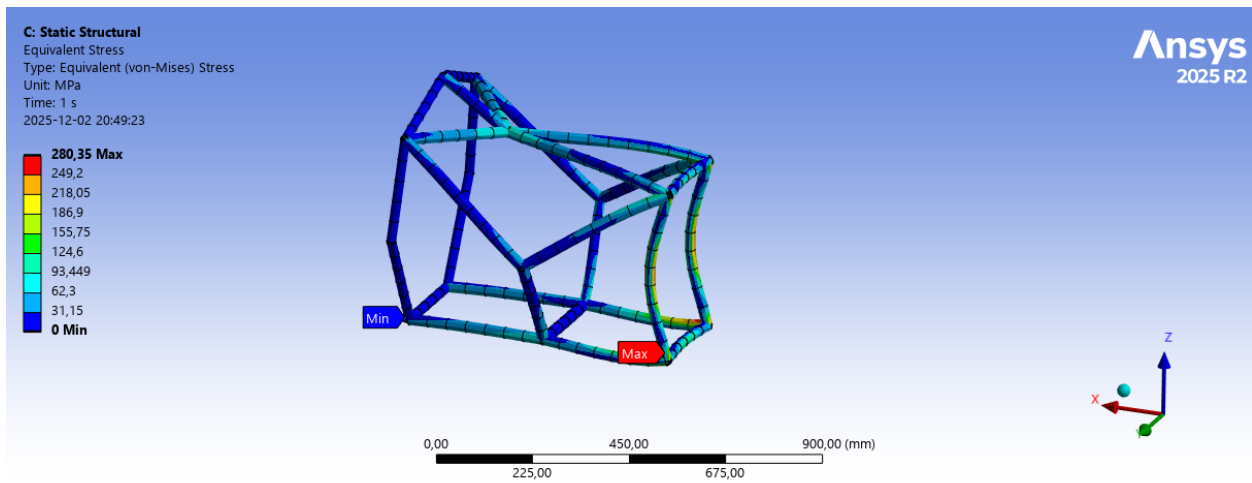


Figure 38: Max stress at plane of impact

The aluminium alloy of cross section OD $\phi 25.4\text{mm}$ and ID $\phi 20.57\text{mm}$ is used in impact member, cross section of OD $\phi 24\text{mm}$ and ID $\phi 20\text{mm}$ is used in the front hoop member and cross section of OD $\phi 28\text{mm}$ and ID $\phi 24\text{mm}$ is used in the rest member of the frame. The total deformation is 5.0029 mm and max stress is 280.35 MPa shown above. The mass of the aluminium alloy frame with different cross sections is 3.7376Kg.

6 Task 6: Gear "&" Rack -revisited

6.1 SubTask: Contacts

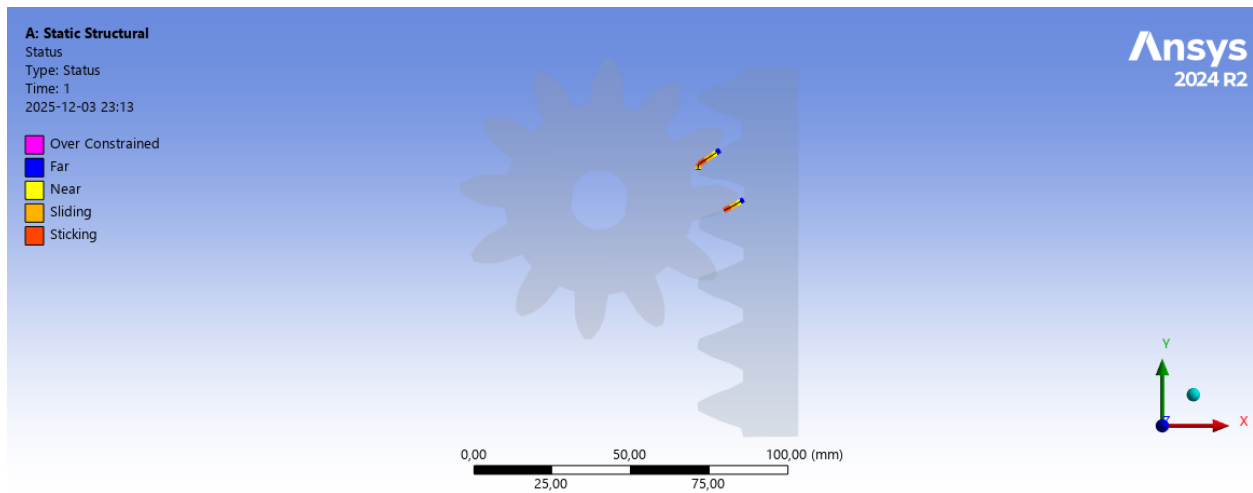


Figure 39: Contact of gear and rack

The contact shows how the gear and rack interact. The contact status changes colour as the gear approaches the rack, moving from far to near and finally to “sliding”, indicating increasing interaction between the tooth surfaces.

6.2 SubTask: Deformation

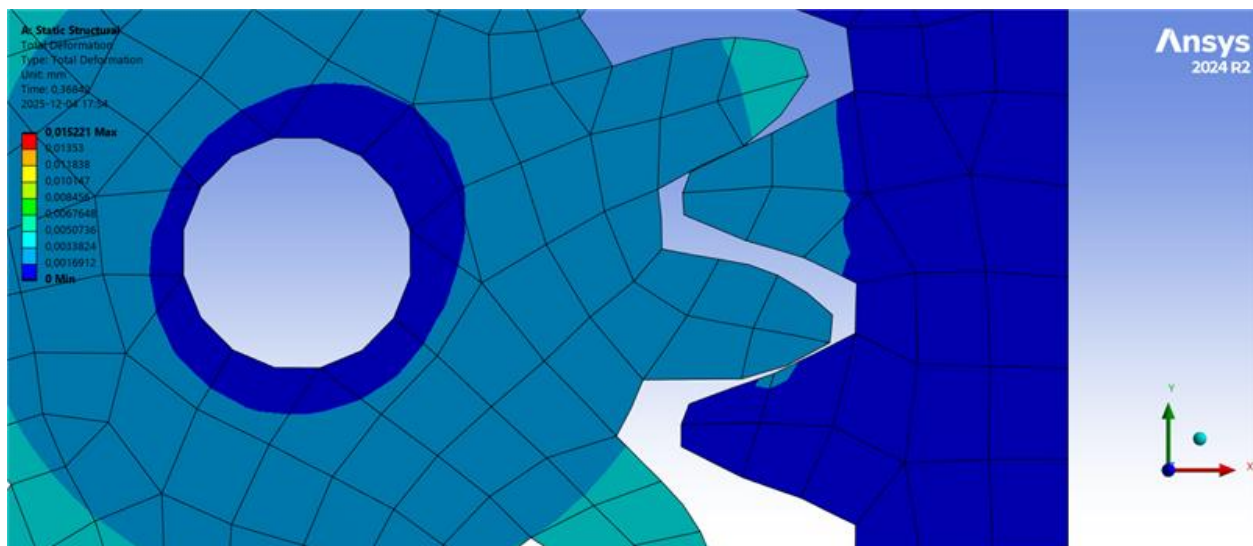


Figure 40: Deformation plot

The deformation starts even though there is initially an air gap between the gear and the rack. This occurs because the rack is fixed while the gear rotates into it, leading to a

nonlinear contact interaction as the teeth come into contact and begin to deform.

6.3 SubTask: Frictionless contact

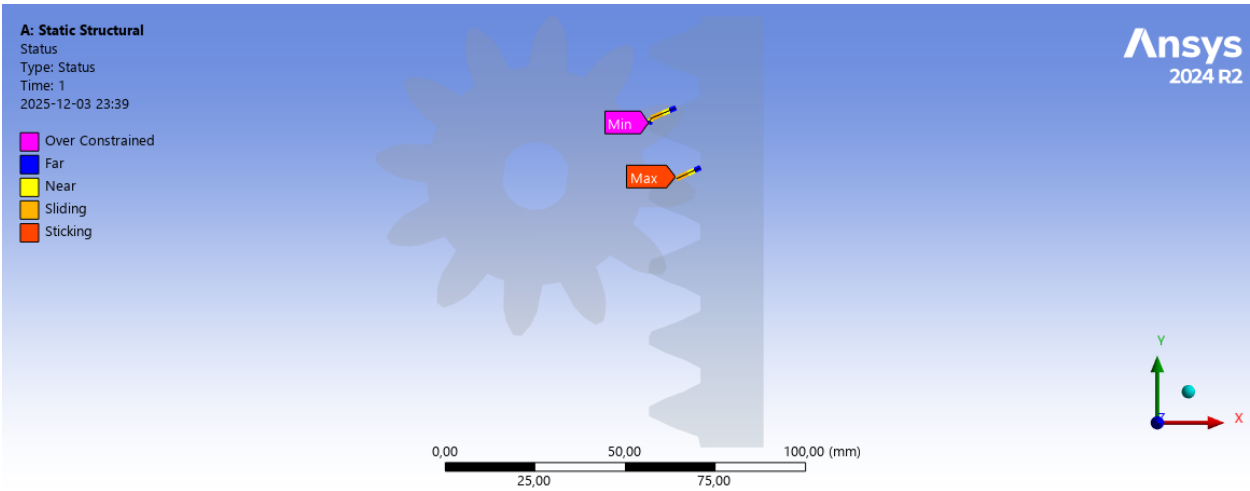


Figure 41: Frictionless contact between gear and rack

When the contact type is changed to frictionless, the gear is able to move freely through the rack. As a result, the sliding contact almost disappears and is mainly replaced by a near contact state, because the system now allows relative motion without significant deformation of the teeth.

6.4 SubTask: Pressure

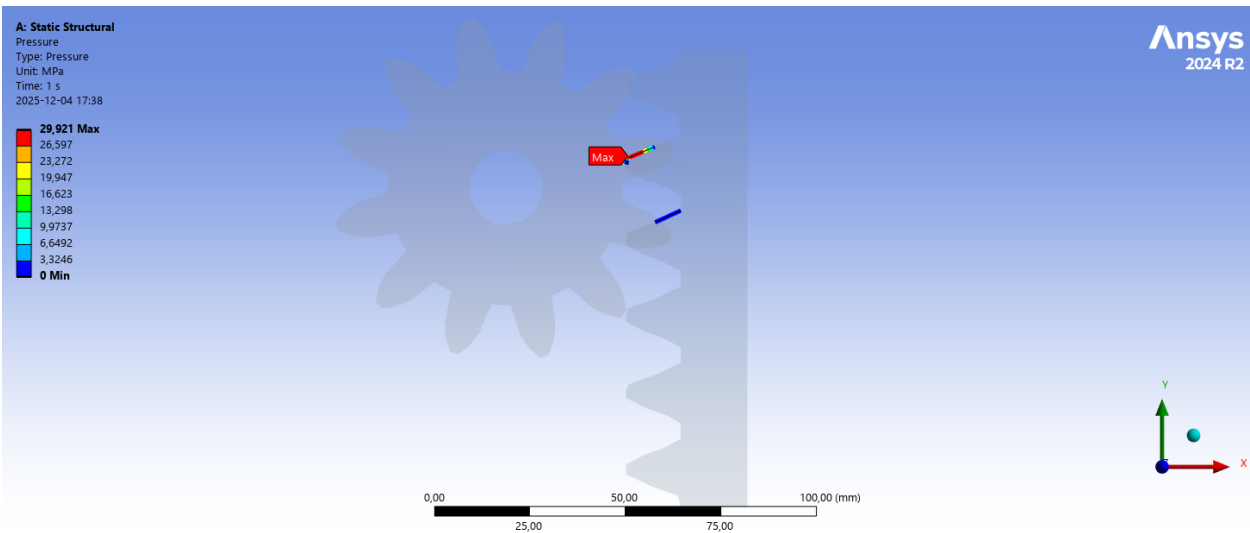


Figure 42: Pressure contact between gear and rack

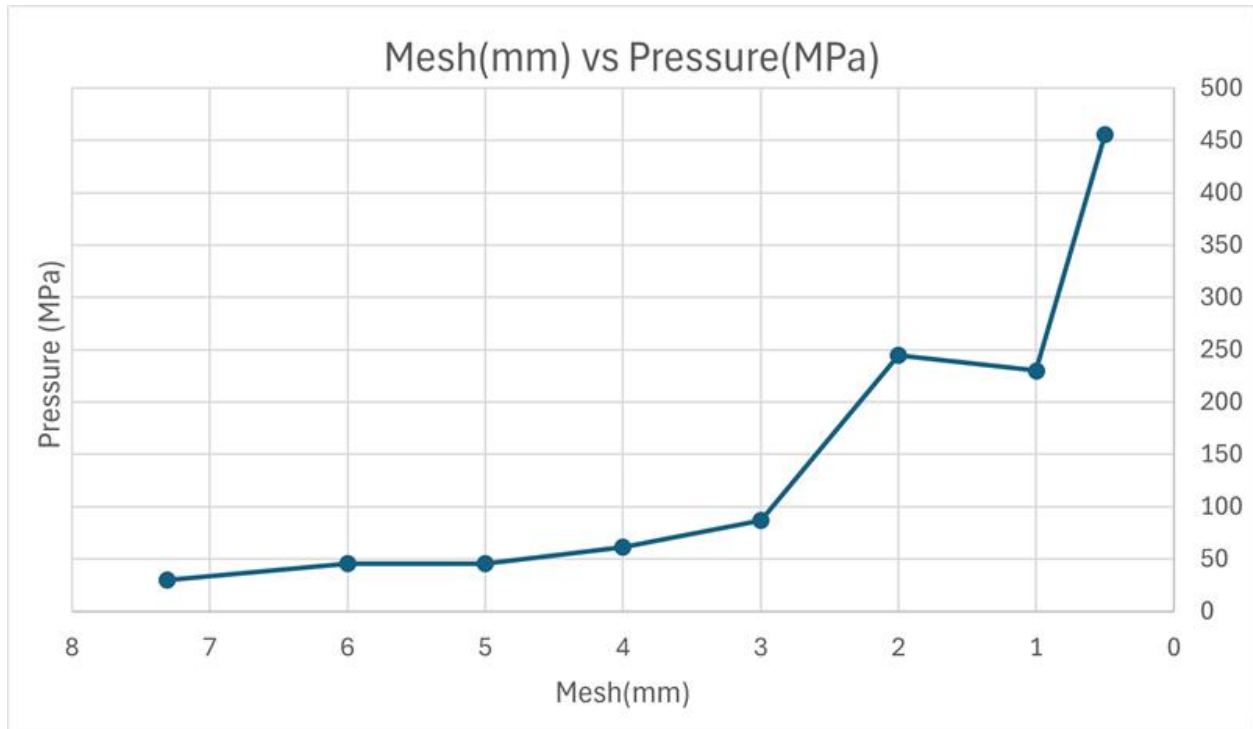


Figure 43: Convergence graph with allowable 5%

Mesh Size(mm)	Pressure(Mpa)	Equivalent stress (Mpa)	%change in pressure
0,5	455,7	445,52	
1	229,8	281,16	-49,5720869
2	244,43	221,26	6,36640557
3	86,496	102,36	-64,61318169
4	61,077	123,39	-29,38748613
5	45,523	95,696	-25,46621478
6	45,503	99,426	-0,043933836
7,3066	29,921	83,994	-34,24389601

Figure 44: Convergence Percentage

This percentage was calculated manually because the convergence feature in Ansys was giving error.

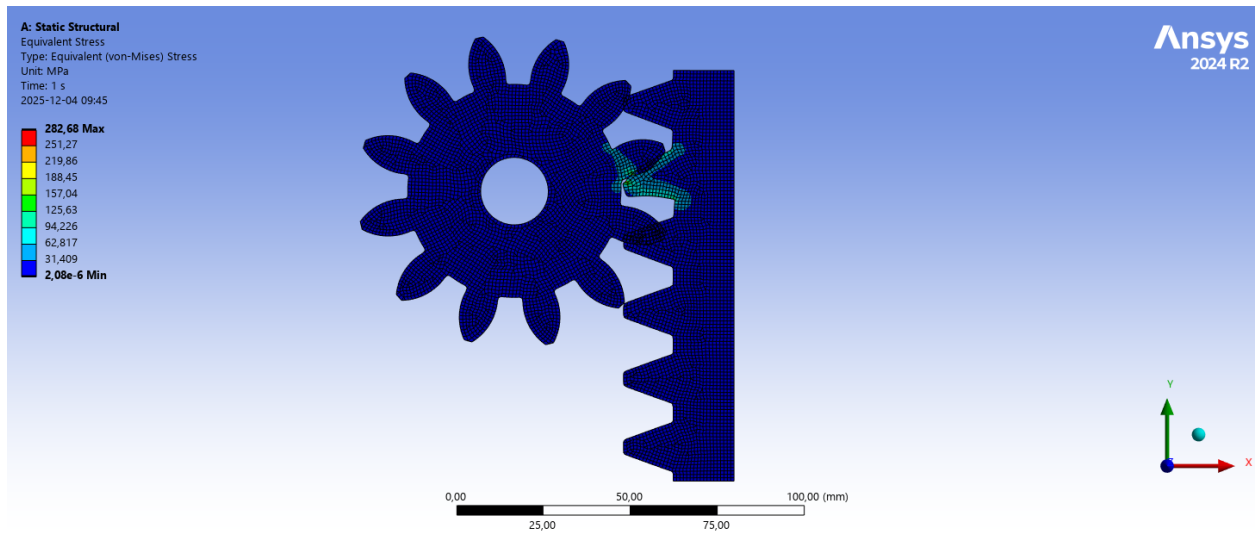


Figure 45: Von mises stress

The von mises stress is around 282.68 MPa at a point near the contact region which is very high.

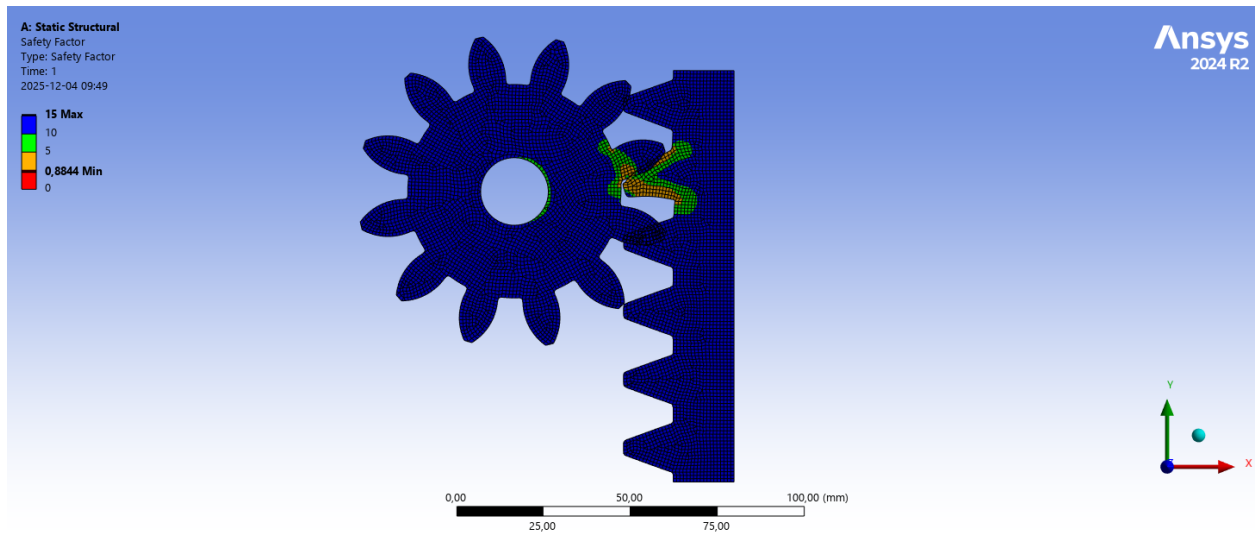


Figure 46: FOS of gear and rack

The minimum FOS obtained was 0.8844, indicating failure and demonstrating that the CA 1 analysis significantly underestimated the stresses. This highlights how crucial it is to define the contact type correctly in order to obtain reliable results.

6.5 SubTask : Normal Lagrange

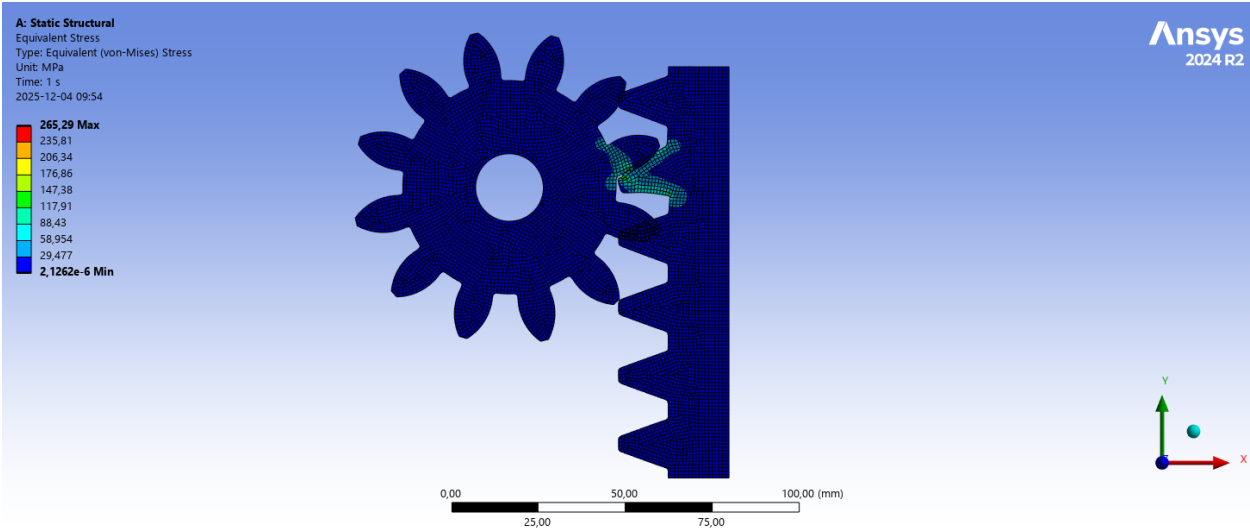


Figure 47: Stress value after Normal Lagrange

This is the von mises stress after normal lagranges and the value is 265.29 Mpa

6.6 SubTask : Force Convergence



Figure 48: Frictionless contact

With frictionless contact, the solution reached force convergence after just two iterations to achieve equilibrium.

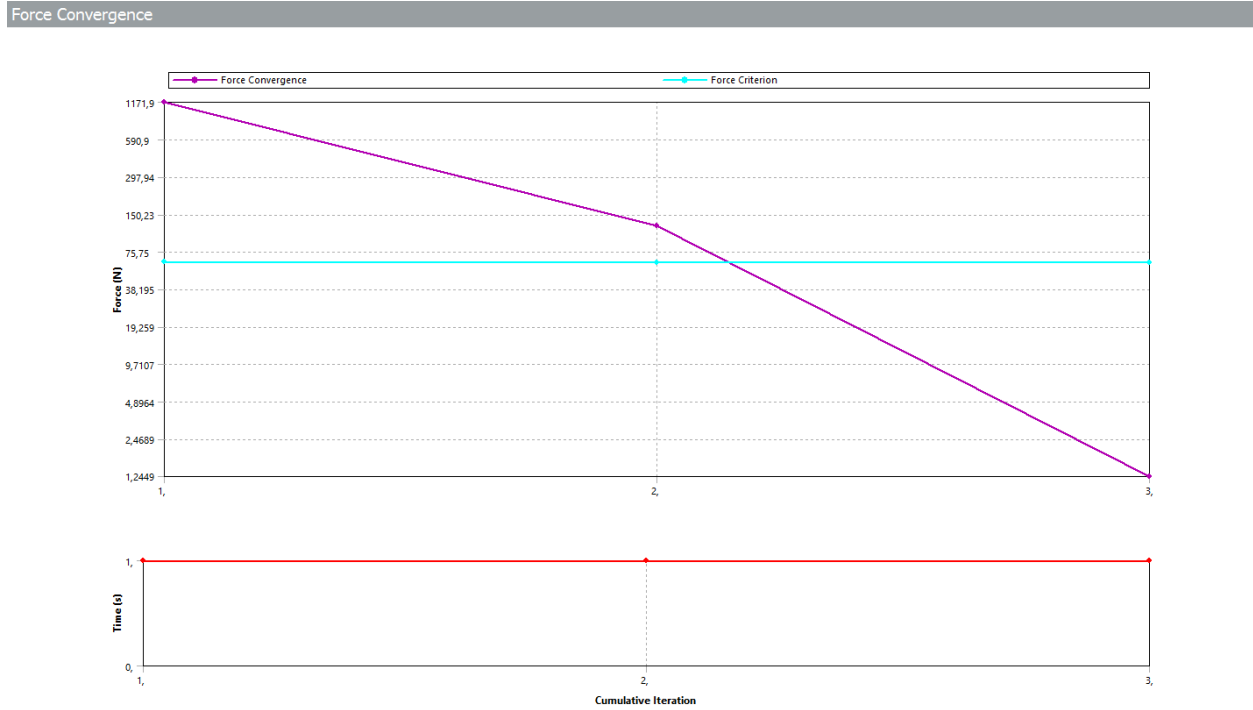


Figure 49: Friction contact

Changing the contact to friction and setting the coefficient of friction to 0.2 With friction contact, the solution reached force convergence after three iterations to achieve equilibrium.

7 Task7: Convergence and divergence

7.1 SubTask: Hole

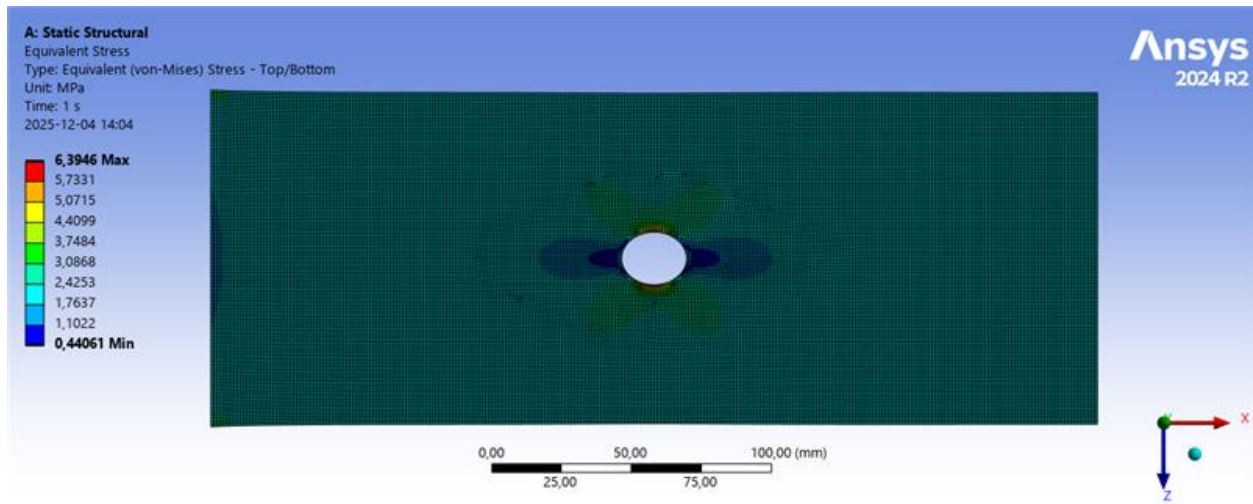


Figure 50: Stress plot

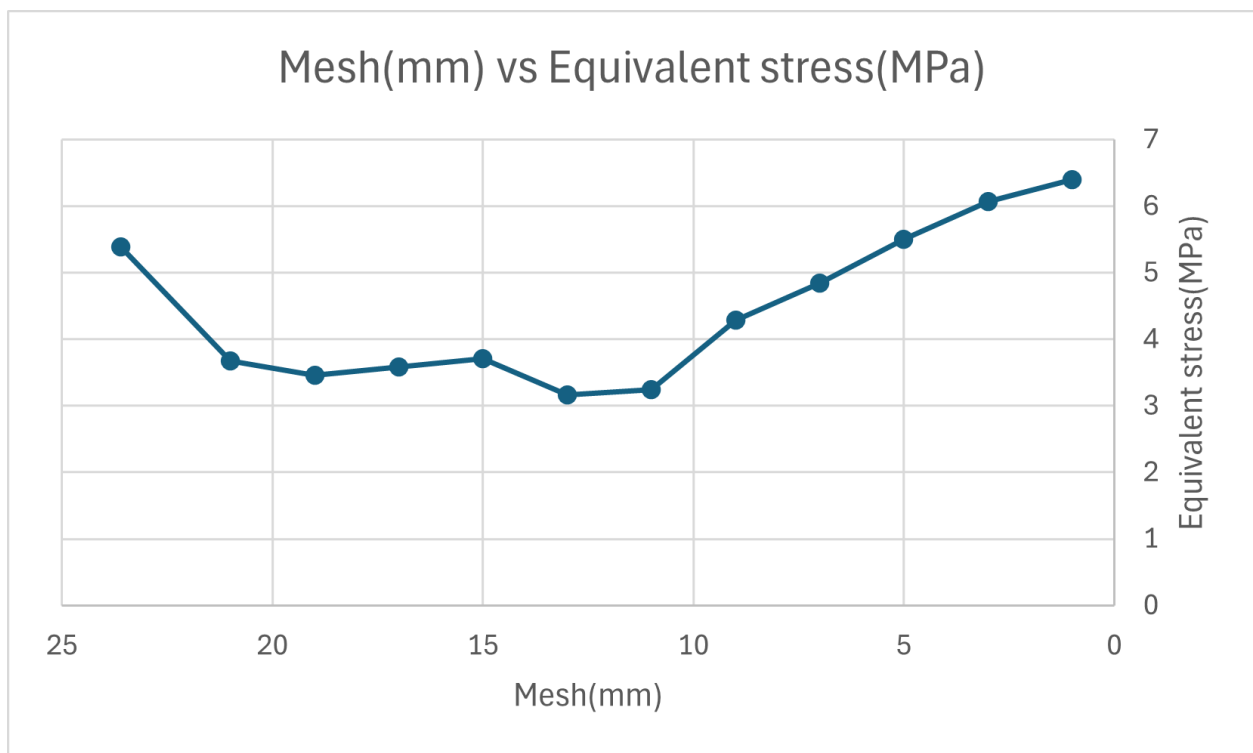


Figure 51: Stress(MPa) vs Mesh(mm)

HOLE		
Mesh(mm)	Equivalent stress(Mpa)	% Percentage convergence
1	6,3946	
3	6,0713	-5,055828355
5	5,505	-9,327491641
7	4,8402	-12,07629428
9	4,2899	-11,3693649
11	3,2374	-24,53437143
13	3,1609	-2,363007352
15	3,7027	17,14068778
17	3,5875	-3,111243147
19	3,4602	-3,548432056
21	3,6766	6,253973759
23,614	5,3844	46,45052494

Figure 52: Percentage convergence of hole

This is manual analysis Stress vs mesh and percentage convergence of hole. For each mesh step, the peak von Mises stress was extracted and added to a stress–vs–mesh-size curve, and convergence was considered.

b) ANSYS convergence is not performed as it is giving error while performing the task.

7.2 SubTask: Square Hole

a) The round hole is replaced by square hole and Stress vs mesh and percentage convergence of square hole is performed. we can see that for mesh 1 mm the convergence is occurred.

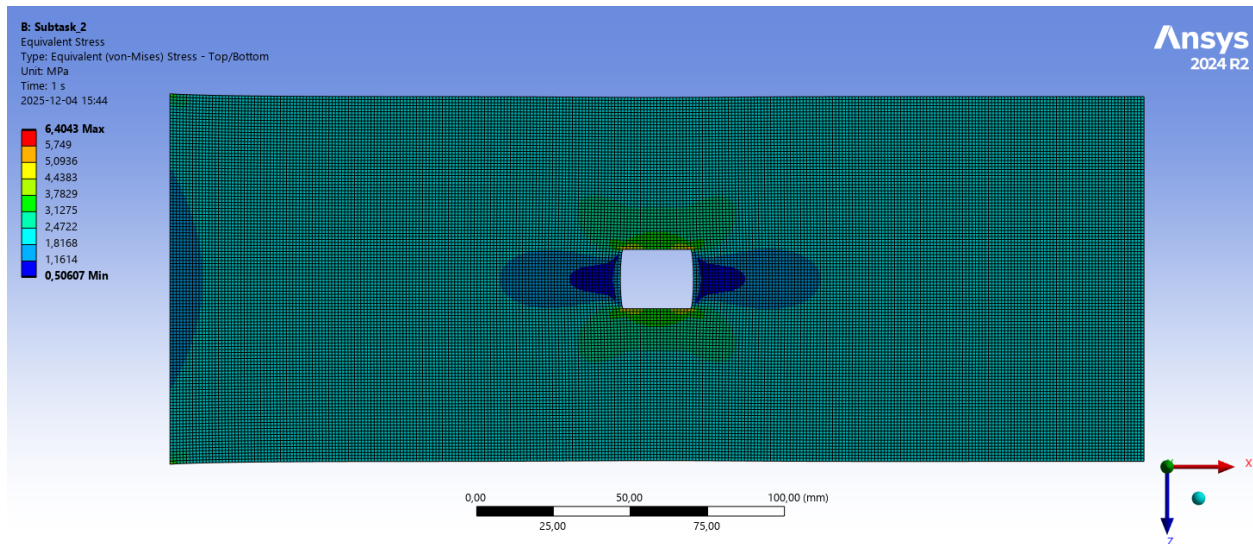


Figure 53: Equivalent stress plot

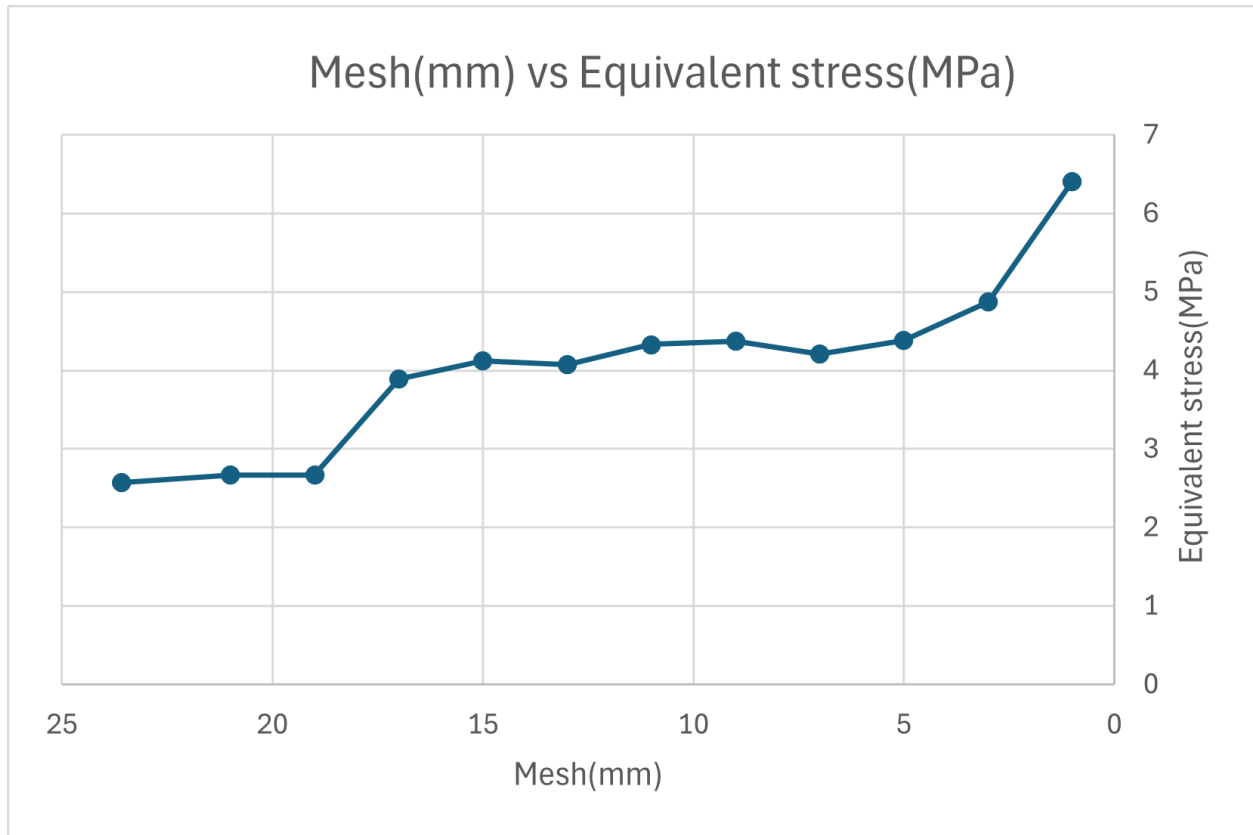


Figure 54: Stress(MPa) vs Mesh(mm)

Square Hole		
Mesh(mm)	Equivalent stress(Mpa)	% Percentage convergence
0,5	7,8663	
1	6,4043	-18,58561204
3	4,8753	-23,87458426
5	4,3855	-10,04656124
7	4,2094	-4,015505644
9	4,372	3,862783295
11	4,3286	-0,992680695
13	4,0689	-5,999630365
15	4,1209	1,277986679
17	3,8876	-5,661384649
19	2,6641	-31,47185924
21	2,6682	0,153898127
23,585	2,5664	-3,815306199

Figure 55: Percentage convergence of Square hole

b)NonLinear Structural Steel

For nonlinear structural steel, the results are similar to the previous problem. For mesh 1 mm the convergence is occurred.

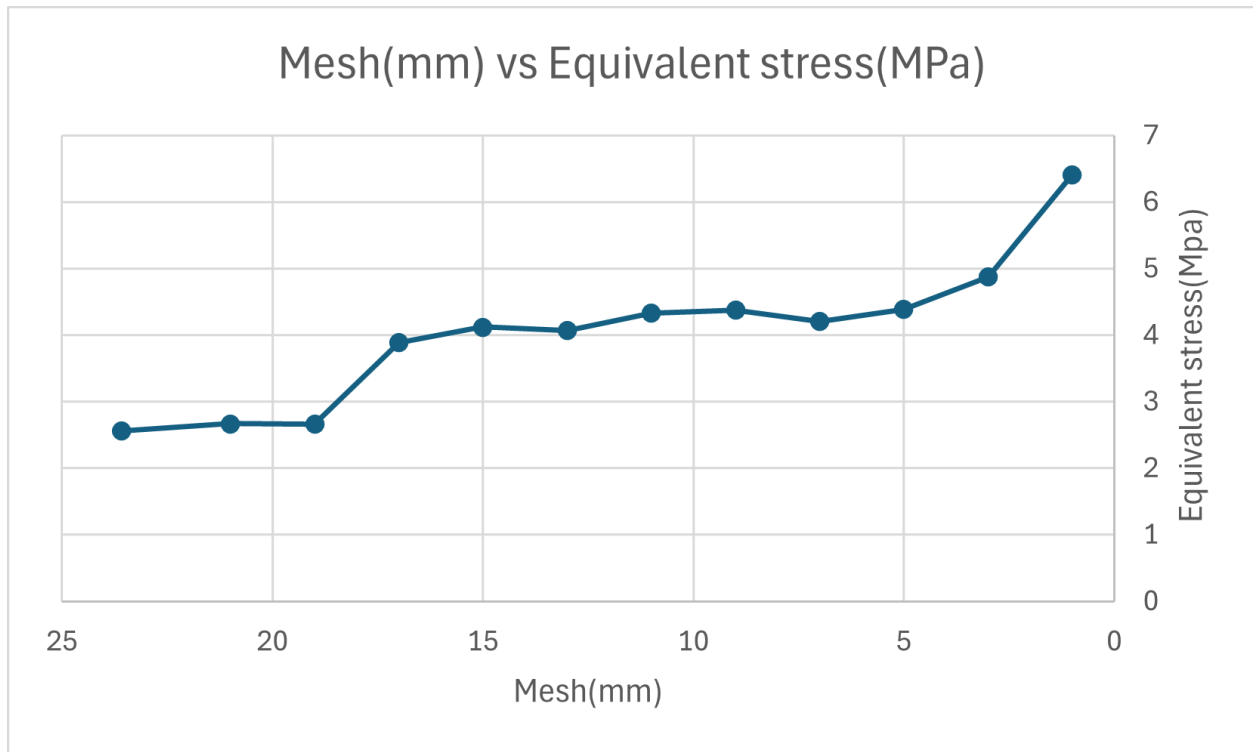


Figure 56: Stress(MPa) vs Mesh(mm)

Square Hole (Structural steel non linear)		
Mesh(mm)	Equivalent stress(Mpa)	% Percentage convergence
0,5	7,8663	
1	6,4043	-18,58561204
3	4,8753	-23,87458426
5	4,3855	-10,04656124
7	4,2094	-4,015505644
9	4,372	3,862783295
11	4,3286	-0,992680695
13	4,0689	-5,999630365
15	4,1209	1,277986679
17	3,8876	-5,661384649
19	2,6641	-31,47185924
21	2,6682	0,153898127
23,585	2,5664	-3,815306199

Figure 57: Percentage convergence of Square hole